
When Shift Happens - Confounding Is to Blame

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Abstract

1 Distribution shifts inject distributional uncertainty into the learning problem and
2 thereby degrade the robustness and generalization of machine learning models.
3 The prevailing hypothesis holds that estimating causal-invariant representations
4 confers robustness to such shifts; recent empirical evidence, however, contradicts
5 this hypothesis on two counts: (i) empirical risk minimization (ERM) attains
6 predictive accuracy comparable to, or exceeding, that of state-of-the-art out-of-
7 distribution (OOD) generalization methods, and (ii) its OOD accuracy increases
8 monotonically as the conditioning set is expanded from causal covariates to the
9 full set of available covariates. Combining empirical measurements with a formal
10 information-theoretic analysis, we identify hidden confounding as the mechanism
11 responsible for this behavior. A distributional shift in an unobserved confounder
12 propagates to the observed data and induces the joint of covariates and target to
13 vary across environments in a manner that violates the invariance assumptions
14 underlying existing OOD generalization methods. Under this regime we prove that
15 maximizing predictive information necessitates estimating environment-specific
16 conditional relationships, so that invariance-based objectives, while necessary,
17 are provably insufficient. We further establish that augmenting the covariate
18 set with proxies for the hidden confounder tightens the achievable predictive
19 information and mitigates the effect of confounding shift. These results yield formal
20 characterizations and actionable design principles for robust OOD generalization
21 algorithms and for principled covariate-selection strategies.

1 Introduction

22 Generalization—the transfer of inferences estimated on an observed sample to a distinct, unobserved
23 population—is a foundational objective across the quantitative sciences. In medicine and the social
24 sciences it is formalized as *external validity*, the requirement that estimates obtained on one population
25 remain valid under a change of population [13]; in ecology it underlies *space-for-time substitution*, in
26 which spatial variation is treated as a surrogate for temporal variation in order to identify long-horizon
27 dynamics [60]; and in engineering it is instantiated as *robust control*, where a controller must satisfy
28 performance specifications under bounded, unmodeled disturbances [35]. Machine learning has
29 correspondingly emerged as a principal methodology for estimating models that generalize across
30 such settings [100, 88, 49].

32 Reliable generalization requires the estimated model to be robust to distribution shift between the
33 training and deployment distributions. Consider, as a running example, a predictor of *food stamp*
34 *reciprocity* from *household attributes* that is estimated on data from one state and deployed in a
35 second state whose underlying distribution differs. A model is said to attain *out-of-distribution (OOD)*
36 *generalization* when its risk remains controlled on both the in-distribution (ID) data—from which the
37 training sample is drawn—and the OOD test data drawn from a shifted distribution. Over the past
38 decade, several method families have been proposed to guarantee OOD generalization under explicit
39 structural assumptions: domain generalization [56, 31, 53, 99, 6, 38, 74], domain adaptation [98,
40 82, 51, 90, 91, 81], and robust learning [42, 69, 94]. Empirically, however, when model selection
41 is performed with care [30, 38, 87], predictors estimated by standard *empirical risk minimization*
42 (*ERM*) [86] achieve OOD accuracy competitive with these specialized methods across a broad range of

real-world applications [30, 24, 50, 57, 67, 87]. A recent controlled study [57] on 16 real-world tabular datasets reports a stronger effect: conditioning the predictor on all available covariates—irrespective of whether a given covariate exerts a direct causal effect on the outcome—yields higher OOD accuracy than conditioning on causal covariates alone. These observations are inconsistent with the invariance-based assumptions that dominate the field and motivate a formal analysis of the mechanisms that govern OOD generalization.

Distribution shift arises whenever the data are generated by a family of distinct environments [71]. In the running example, the distribution of *food stamp reciprocity* varies across states because each state imposes distinct eligibility rules, so each state indexes a separate environment-specific distribution. Within such a family, a subset of the statistical relationships between covariates and outcome may be preserved across every environment; these are termed *invariant relationships* [6]. Identifying and estimating the invariant relationships is the standard route to provable OOD generalization [6, 59, 31, 66, 56, 98, 63]. The strongest such condition is *causal invariance*, under which the causal mechanisms—and hence the conditional distribution of the outcome given its causal parents—remain fixed across environments [71, 59, 6]. In practice, however, the covariate set required to predict the outcome is only partially observed, owing to constraints on data collection, privacy restrictions, or measurement error [14, 52, 20, 48]. Omitting these variables breaks the invariance that generalization relies on. The failure is more severe when the unobserved variables are confounders that act simultaneously on the observed covariates and on the outcome. Such hidden confounders are pervasive, and a shift in their marginal distribution is precisely what distinguishes one environment from another. Failing to model these shifts not only degrades generalization [40, 4, 85, 62] but can induce the estimator to recover a relationship between the observed covariates and the outcome whose sign is opposite to the true one.

Despite its practical significance, the effect of hidden confounding shift on OOD generalization has received limited formal treatment [4, 40, 85, 62]. This work supplies a theoretical characterization, corroborated by empirical measurements, that accounts for the recently reported findings [30, 24, 50, 57, 67, 87]. Our contributions are as follows:

- We formalize confounding shift within a structural causal framework (§ 3) and derive why enlarging the conditioning set with informative, non-causal covariates can strictly increase attainable predictive accuracy.
- We prove that maximizing the predictive information between predictions and outcomes requires explicitly estimating environment-specific conditional relationships; hence invariant representations, though necessary, are provably insufficient (§ 4.1, § 4.2).
- We show that any covariate informative of the outcome or of the hidden confounder raises the attainable predictive information between predictions and outcomes, thereby providing a formal criterion for covariate selection under hidden confounding shift (§ 4.3).
- Through experiments on real-world and synthetic datasets we provide evidence that (i) hidden confounding shift is detectable and prevalent in real-world tabular benchmarks, (ii) the estimated degree of environment-specific structure is positively correlated with both ID and OOD test accuracy, and (iii) augmenting the conditioning set with informative, non-causal covariates increases OOD accuracy (§ 5).

Taken together, our analysis accounts for the empirically observed competitiveness of ERM under real-world distribution shift and quantifies the systematic, and largely unavoidable, role of hidden confounding in OOD generalization, thereby motivating estimators that address it explicitly.

2 Related work

Out-of-distribution (OOD) generalization subsumes several distinct problem formulations, each defined by the information available at training and test time: domain generalization [56, 31, 53, 99, 6, 38], domain adaptation [98, 82, 51, 90, 91, 81], robust learning [42, 69, 94], federated learning [54, 36, 44, 97, 45], and OOD detection [41, 32, 92, 84, 49]. Domain adaptation presupposes access to unlabeled samples from the test distribution; domain generalization instead assumes observed environment labels at training time; federated learning operates under a distributed protocol subject to communication-efficiency and privacy constraints when the sample is partitioned across environments; and OOD detection reduces to a hypothesis test that flags samples deviating from the training distribution. A shared objective across these formulations is the estimation of invariant

relationships. Recent analyses, however, establish that inductive biases beyond invariance are required to attain the optimal risk [46, 72, 93]. Ye et al. [93] prove that feature invariance is necessary but not sufficient for generalization and formalize the complementary role of feature *informativeness*. Our analysis characterizes invariance and informativeness jointly, as two terms in a single decomposition of predictive information.

Proxy variable adjustment: For the setting in which the confounder is observed during training but unobserved at test time, Landeiro and Culotta [40] derive an adjustment for the confounding shift that provably improves classifier risk. Extending proxy-based identification results for causal effect estimation [55, 39], subsequent domain adaptation methods recast estimation under unknown shift as a causal-effect identification problem [4, 85]. OOD generalization under hidden confounding shift has been analyzed under an overlapping-support condition on the confounder [62]. Rather than proposing a further adjustment procedure, we characterize *the mechanism by which* proxy variables raise the attainable predictive information, and thereby explain when and why they improve OOD generalization.

All-covariate models versus causal models. Nastl and Hardt [57] construct a benchmark in which covariates are partitioned into four classes—causal (conservatively labeled), arguably causal, anti-causal, and otherwise spurious—and show that, across 16 datasets, the predictor conditioned on all covariates Pareto-dominates predictors conditioned on the causal or arguably-causal subsets, jointly on ID and OOD accuracy. A formal account of this dominance is largely absent from the literature; we supply the structural conditions and the corresponding argument that reproduce it. For linear structural causal models, anchor regression [68] parameterizes a continuum between two estimators—one conditioning on all observed covariates and one conditioning on causal covariates alone. Our objective is to characterize, under hidden confounding shift, the effect on attainable accuracy of expanding the conditioning set with non-causal covariates. Eastwood et al. [21] prove that unstable covariates can lower risk when they are informative of the label and conditionally independent of the stable covariates given the label, and propose an adjustment estimated from unlabeled test-domain data. Yet on a medical dataset not constructed for this task [8], ERM remains competitive with their adjustment, consistent with Nastl and Hardt [57]. This is a manifestation of the general result that, under a correctly specified covariate shift, maximum likelihood estimation (MLE) is minimax optimal for OOD generalization [26]. Real-world settings, however, rarely satisfy the correct-specification condition, precisely because of hidden confounding shift—the regime this work analyzes.

3 Manifestations of hidden confounding shift

This section formalizes hidden confounding shift and establishes, via an identification argument for the causal effect of the observed covariates on the outcome, why it must be modeled explicitly.

Taxonomy of distribution shifts. For a covariate vector $\mathbf{X}$ and target Y , a pair of environments e and e' can differ in any of the marginal or conditional distributions enumerated in Table 1. Each such shift is, in general, the downstream consequence of a shift in the marginal $\mathbb{P}(U)$ of an unobserved covariate U that acts on $\mathbf{X}$, on Y , or on both (Figure 1). Marginalizing over U makes the propagation explicit: since $\mathbb{P}(\mathbf{X}) = \sum_U \mathbb{P}(U)\mathbb{P}(\mathbf{X} | U)$, whenever $\mathbb{P}^e(U) \neq \mathbb{P}^{e'}(U)$ and $U \rightarrow \mathbf{X}$, it follows that $\mathbb{P}^e(\mathbf{X}) \neq \mathbb{P}^{e'}(\mathbf{X})$. The same marginalization argument transfers verbatim to $\mathbb{P}(Y)$, $\mathbb{P}(\mathbf{X} | Y)$, $\mathbb{P}(Y | \mathbf{X})$, and $\mathbb{P}(\mathbf{X}, Y)$.

Existing OOD generalization methods impose specific causal relationships among U , $\mathbf{X}$, Y , each of which entails a particular invariance. Under $U \rightarrow Y \rightarrow \mathbf{X}$ (Figure 1 (a)), the factorization $\mathbb{P}(\mathbf{X}, Y) = \mathbb{P}(Y)\mathbb{P}(\mathbf{X} | Y)$ implies that any joint shift $\mathbb{P}^e(\mathbf{X}, Y) \neq \mathbb{P}^{e'}(\mathbf{X}, Y)$ is attributable to label shift, $\mathbb{P}^e(Y) \neq \mathbb{P}^{e'}(Y)$, while the conditional $\mathbb{P}(\mathbf{X} | Y)$ is invariant, $\mathbb{P}^e(\mathbf{X} | Y) = \mathbb{P}^{e'}(\mathbf{X} | Y)$, since $\mathbb{P}(\mathbf{X} | Y, U) = \mathbb{P}(\mathbf{X} |$

Table 1: Taxonomy of distribution shifts, each defined by the pair of environment-conditional distributions it renders unequal.

Type of Shift	Mathematical Expression
Label	$\mathbb{P}^e(Y) \neq \mathbb{P}^{e'}(Y)$
Covariate	$\mathbb{P}^e(\mathbf{X}) \neq \mathbb{P}^{e'}(\mathbf{X})$
Conditional Covariate	$\mathbb{P}^e(\mathbf{X} Y) \neq \mathbb{P}^{e'}(\mathbf{X} Y)$
Concept	$\mathbb{P}^e(Y \mathbf{X}) \neq \mathbb{P}^{e'}(Y \mathbf{X})$
	$\mathbb{P}^e(\mathbf{X}, Y) \neq \mathbb{P}^{e'}(\mathbf{X}, Y)$

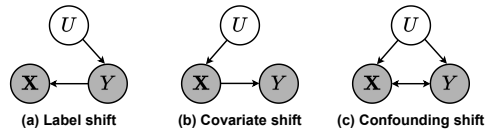


Figure 1: Causal graphs underlying distribution shifts.

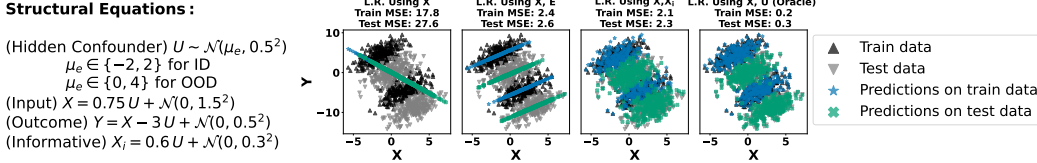


Figure 2: Four linear regression (L.R.) estimators evaluated under hidden confounding shift with minimal environment overlap (widely separated μ_e). (i) Conditioning on X alone recovers a relationship with Y of the wrong sign—Simpson’s paradox. (ii) Conditioning additionally on environment-specific summary statistics E of X restores the correct sign but is constrained by the representational capacity of E . (iii) Conditioning on an informative covariate X_i that proxies U further reduces OOD error. (iv) The oracle conditions on X and U and attains the minimum achievable error.

152 Y) [89, 83, 25, 5, 47, 96, 18, 79]. Symmetrically, under $U \rightarrow \mathbf{X} \rightarrow Y$ (Figure 1 (b)), the joint
153 shift is attributable to covariate shift, $\mathbb{P}^e(\mathbf{X}) \neq \mathbb{P}^{e'}(\mathbf{X})$, while the predictive conditional is invariant,
154 $\mathbb{P}^e(Y | \mathbf{X}) = \mathbb{P}^{e'}(Y | \mathbf{X})$ [70, 16, 80, 11, 28]. Neither invariance holds in the generic case in which
155 U acts on both $\mathbf{X}$ and Y (Figure 1 (c)) [50, 4, 40, 85, 62, 65, 64]: a shift in $\mathbb{P}(U)$ then induces
156 label shift, covariate shift, conditional covariate shift, and concept shift simultaneously. The only
157 invariance that survives is the confounder-conditional joint, $\mathbb{P}^e(\mathbf{X}, Y | U) = \mathbb{P}^{e'}(\mathbf{X}, Y | U)$ —the
158 *stable confounding shift* condition [85, 4]—and, because U is unobserved, this invariance is not
159 identifiable from the observed data.

160 **Information-theoretic quantification of shift:** Following Federici et al. [22], we quantify each shift
161 by the mutual information between the corresponding random variables and the environment index
162 E . Concretely, $I(\mathbf{X}; E)$, $I(Y; E)$, $I(\mathbf{X}; E|Y)$, $I(Y; E|\mathbf{X})$, and $I(\mathbf{X}, Y; E)$ measure, respectively,
163 the shifts in $\mathbb{P}(\mathbf{X})$, $\mathbb{P}(Y)$, $\mathbb{P}(\mathbf{X} | Y)$, $\mathbb{P}(Y | \mathbf{X})$, and $\mathbb{P}(\mathbf{X}, Y)$ across environments; each mutual-
164 information term is zero if and only if the associated distribution is environment-invariant. In these
165 terms, the label-shift assumption is the constraint $I(\mathbf{X}; E | Y) = 0$ and the covariate-shift assumption
166 is the constraint $I(Y; E | \mathbf{X}) = 0$. Under hidden confounding shift, as established above, both
167 quantities are generically strictly positive.

168 **A diagnostic instance of hidden confounding shift:** The experiment in Figure 2 isolates the failure
169 mode. The outcome Y is generated from the observed covariate X and a hidden confounder U , and
170 environments are indexed by the environment-specific mean μ_e of U . A linear regression estimated
171 on X alone recovers a relationship between X and Y of the wrong sign (first subplot): although Y
172 increases with X within every environment, the pooled estimator returns a decreasing fit—an instance
173 of Simpson’s paradox [73]. The invariance-based estimators IRM [6], VREX [38], and Group
174 DRO [69] exhibit the same sign reversal. We prove in § 4 that, under this regime, the risk-minimizing
175 strategy is to estimate environment-specific conditional relationships, which restores the correct sign
176 of the X – Y relationship. The second subplot confirms this: conditioning on environment-specific
177 summary statistics of X (mean, standard deviation, and quantiles) recovers the within-environment
178 relationship. This is the information-theoretic analogue of *backdoor adjustment* [58], with the
179 environment-specific statistics serving as an adjustment set that d-separates the confounding path.
180 The representational capacity of statistics derived from X alone is nonetheless limited, so the recovery
181 in the second subplot is only partial. As we show in § 4.3, augmenting the conditioning set with
182 informative covariates that proxy the unobserved confounder yields a further reduction in error (third
183 subplot). The oracle regression conditioned on (X, U) is included as the Bayes-optimal reference
184 and attains the minimum achievable mean squared error.

185 4 Impact of hidden confounding shifts on OOD generalization

186 This section develops a formal analysis of how hidden confounding shift affects the constituent
187 quantities that govern OOD generalization. We write the predictor as the composition $\hat{Y} = (f \circ \phi)(\mathbf{X})$,
188 in which ϕ is a representation (feature) map and f is a classifier acting on $\phi(\mathbf{X})$. Entropy and
189 mutual information are defined as $H(X) = -\mathbb{E}_X[\log(\mathbb{P}(X))]$ and $I(X; Y) = \mathbb{E}_{X,Y}[\log \frac{\mathbb{P}(X,Y)}{\mathbb{P}(X)\mathbb{P}(Y)}]$,
190 respectively. The population risk of $(f \circ \phi)$ in environment e is $\mathcal{R}^e(f \circ \phi) = \mathbb{E}_{(\mathbf{x}, Y) \sim \mathbb{P}_{\mathbf{x}, Y}^e}[\ell((f \circ \phi)(\mathbf{x}), Y)]$ for a loss $\ell(\cdot, \cdot)$. OOD generalization is the problem of estimating $(f \circ \phi)$ with low
191 risk on both ID and OOD distributions, and the literature instantiates it through distinct objectives.
192 Robust-optimization methods [7, 9, 19, 75, 69] minimize the worst-case training risk $\mathcal{R}^{\text{Rob}}(f \circ \phi) =$
193 $\max_{e \in \mathcal{E}_{tr}} \mathcal{R}^e(f \circ \phi)$ over the training environments $\mathcal{E}_{tr}$. Variance risk extrapolation (VREX) [38]

minimizes the mean training risk penalized by its cross-environment variance, $\mathcal{R}^{\text{VREX}}(f \circ \phi) = \sum_{e \in \mathcal{E}_{tr}} \mathcal{R}^e(f \circ \phi) + \beta \times \text{Variance}(\{\mathcal{R}^1(f \circ \phi), \dots, \mathcal{R}^{|\mathcal{E}_{tr}|}(f \circ \phi)\})$ with penalty weight β . Invariant risk minimization (IRM) [6] minimizes $\sum_{e \in \mathcal{E}_{tr}} \mathcal{R}^e(f \circ \phi)$ subject to the constraint that a single f is simultaneously risk-optimal in every environment. Empirical risk minimization (ERM) pools the samples across all training environments and minimizes the aggregate empirical risk $\mathcal{R}^{\text{ERM}}(f \circ \phi) = \sum_{e \in \mathcal{E}_{tr}} \sum_{(\mathbf{X}, Y) \in \mathcal{D}^e} \ell((f \circ \phi)(\mathbf{X}), Y)$, where $\mathcal{D}^e$ is the sample observed in environment e [6, 38].

To expose how hidden confounding shift acts on these objectives, we adopt the predictive information $I(Y; \hat{Y})$, with $\hat{Y} = (f \circ \phi)(\mathbf{X})$, as the unifying quantity of interest. $I(Y; \hat{Y})$ measures the reduction in uncertainty about the label Y afforded by the prediction $\hat{Y}$, and thus serves as a task-agnostic surrogate for predictive performance. We first formalize two properties of the representation $\phi(\mathbf{X})$ —*informativeness* and *invariance*—that jointly determine OOD generalization [93].

Definition 4.1 (Informativeness and Conditional Informativeness). *The informativeness of features $\phi(\mathbf{X})$ for predicting Y is defined as $\text{INF}(\phi(\mathbf{X}), Y) = I(\phi(\mathbf{X}); Y)$. The conditional informativeness of features $\phi(\mathbf{X})$ for predicting Y conditioned on environment variable E is defined as $\text{CINF}(\phi(\mathbf{X}), Y, E) = I(\phi(\mathbf{X}); Y | E)$.*

Conditional informativeness is the information that $\phi(\mathbf{X})$ carries about Y after conditioning on the environment, i.e., the within-environment predictive content. Decreasing CINF discards label-relevant signal, whereas increasing it is contraindicated in settings such as algorithmic fairness, where $\phi(\mathbf{X})$ may encode sensitive within-environment attributes and yield discriminatory predictions. The fairness constraint can be enforced by minimizing $I(\phi(\mathbf{X}); E)$, but at a cost in predictive performance, as we quantify in § 4.2. Absent any sensitive information in E , we treat the maximization of CINF as a legitimate objective for generalization, a position we formalize in § 4.1.

Definition 4.2 (Variation and Invariance). *For a given label Y , the variation in the features $\phi(\mathbf{X})$ across environments E is defined as $\text{VAR}(\phi(\mathbf{X}), Y, E) = I(\phi(\mathbf{X}); E | Y)$. For a given label Y , the invariance of the features $\phi(\mathbf{X})$ across environments E is defined as the negative of the variation i.e., $\text{INV}(\phi(\mathbf{X}), Y, E) = -\text{VAR}(\phi(\mathbf{X}), Y, E) = -I(\phi(\mathbf{X}); E | Y)$.*

Invariance measures the constancy of the representation $\phi(\mathbf{X})$ across environments conditional on Y . Decreasing invariance retains environment-specific structure in $\phi(\mathbf{X})$ and admits overfitting to individual environments, whereas increasing it removes environment-specific dependence and enforces invariant estimation. Continuing the mutual-information parameterization of § 3, we denote by $I(\phi(\mathbf{X}); E)$ the *feature shift* and by $I(Y; E | \phi(\mathbf{X}))$ the *concept shift*. Where the distinction is immaterial, *concept shift* refers interchangeably to $I(Y; E | \mathbf{X})$ or $I(Y; E | \phi(\mathbf{X}))$.

4.1 A general decomposition of predictive information

We first fix the graphical notation and d-separation machinery used throughout. A causal graph $\mathcal{G}$ is a directed graph whose nodes are random variables and whose directed edges denote direct causal influence. A *path* between nodes X_i and X_j is a sequence of distinct nodes joined consecutively by edges. A *directed path* from X_i to X_j ($i < j$) is a path whose edges all orient toward X_j , i.e. $X_i \rightarrow X_{i+1} \rightarrow \dots \rightarrow X_{j-1} \rightarrow X_j$; along it, X_i is the *parent* of X_{i+1} , X_{i+1} the *child* of X_i , X_i an *ancestor* of X_j , and X_j a *descendant* of X_i . Every path is composed of three elementary connections: the *chain* $X_i \rightarrow X_j \rightarrow X_k$, the *fork* $X_i \leftarrow X_j \rightarrow X_k$, and the *collider* $X_i \rightarrow X_j \leftarrow X_k$. For chains and forks the endpoints are marginally dependent and become conditionally independent given the middle node, $X_i \not\perp\!\!\!\perp X_k$ and $X_i \perp\!\!\!\perp X_k | X_j$; for the collider the endpoints are marginally independent and become conditionally dependent given the collider or any of its descendants, $X_i \perp\!\!\!\perp X_k$ and $X_i \not\perp\!\!\!\perp X_k | X_j$. A path is *blocked* by a conditioning set $\mathcal{S}$ if and only if (i) it contains a chain or fork whose middle node lies in $\mathcal{S}$, or (ii) it contains a collider such that neither the collider nor any descendant of it lies in $\mathcal{S}$; otherwise the path is *open*. When $\mathcal{S}$ blocks every path between X_i and X_k , the d-separation criterion yields the conditional independence $X_i \perp\!\!\!\perp X_k | \mathcal{S}$.

We now analyze the predictive information $I(Y; \hat{Y})$ for a prediction $\hat{Y} = (f \circ \phi)(\mathbf{X})$ computed from the representation $\phi(\mathbf{X})$. The joint causal structure over $\mathbf{X}, Y, U, \phi(\mathbf{X}), \hat{Y}$, and E is specified in Figure 3, where U is the hidden confounder and E is the environment index whose value determines $\mathbb{P}(U)$ —so any two distinct environments differ in $\mathbb{P}(U)$. Because $\hat{Y}$ is a

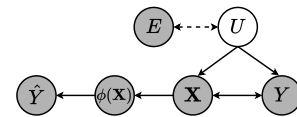


Figure 3: The bi-directed edge between $\mathbf{X}$ and Y encodes that a subset of the covariates in $\mathbf{X}$ are causes of Y while the remainder are effects of Y .

deterministic function of $\phi(\mathbf{X})$, it carries no information about Y beyond $\mathbf{X}$ or $\phi(\mathbf{X})$, giving the conditional independences $\hat{Y} \perp\!\!\!\perp Y \mid \mathbf{X}$ and $\hat{Y} \perp\!\!\!\perp Y \mid \phi(\mathbf{X})$. We first derive a decomposition of the predictive information that is agnostic to the specific mechanism by which U acts on $\mathbf{X}$ and Y ; the mechanism-specific refinement is deferred to § 4.2.

Proposition 4.1. *Let $\mathbf{X}$ be a covariate vector and Y a label with causal structure $\mathbf{X} \leftrightarrow Y$ (a subset of the covariates cause Y and the remainder are caused by Y), let E be the environment variable, ϕ a representation map, and $\hat{Y}$ the prediction. Then the predictive information $I(Y; \hat{Y})$ admits the exact decomposition*

$$I(Y; \hat{Y}) = \overbrace{I(\phi(\mathbf{X}); Y|E)}^{\text{Cond. informativ.}} - \overbrace{\frac{I(\phi(\mathbf{X}); E|Y)}{2}}^{\text{Variation}} + \overbrace{\frac{I(Y; E)}{2}}^{\text{Label shift}} + \overbrace{\frac{I(\phi(\mathbf{X}); E)}{2}}^{\text{Feature shift}} - \overbrace{\frac{I(Y; E|\phi(\mathbf{X}))}{2}}^{\text{Concept shift}} - \overbrace{I(\phi(\mathbf{X}); Y|\hat{Y})}^{\text{Residual}} \quad (1)$$

where the residual $I(\phi(\mathbf{X}); Y|\hat{Y})$ is the label-relevant information in $\phi(\mathbf{X})$ that the prediction $\hat{Y}$ fails to transmit. The identity holds verbatim with $\mathbf{X}$ substituted for $\phi(\mathbf{X})$.

Proofs are deferred to Appendix § A. The decomposition provides a common coordinate system in which existing methods are seen to maximize or minimize specific terms of Equation (1). IRM [6] constrains a single classifier f to be optimal across environments; this is equivalent to maximizing $-I(\phi(\mathbf{X}); E|Y)$, since any residual environment-specific information in $\phi(\mathbf{X})$ conditional on Y would render the shared f suboptimal in some environment. DANN [23] and the independence criterion for fair classification [22] instead minimize the feature shift $I(\phi(\mathbf{X}); E)$, aligning the marginal representation across environments, as do domain-invariant feature methods in domain adaptation [82, 90]. Although minimizing $I(\phi(\mathbf{X}); E)$ serves fairness, it is not risk-optimal [98, 33, 22]: Equation (1) shows directly that reducing $I(\phi(\mathbf{X}); E)$ lowers $I(Y; \hat{Y})$. CDAN [51] enforces the joint alignment $\mathbb{P}^e(\phi(\mathbf{X}), Y) = \mathbb{P}^{e'}(\phi(\mathbf{X}), Y)$, which makes E independent of the pair $(\phi(\mathbf{X}), Y)$; consequently $\mathbb{P}(Y | \phi(\mathbf{X}), E) = \mathbb{P}(Y | \phi(\mathbf{X}))$ and hence $I(Y; E | \phi(\mathbf{X})) = 0$. Thus CDAN drives the concept-shift term to zero implicitly, without including it in its objective. Finally, although the label-shift term $I(Y; E)$ is constant with respect to ϕ and f , methods robust to label distribution shift address it explicitly [69, 95].

Equation (1) is agnostic to the mechanism relating U to $\mathbf{X}$ and Y ; specifying that mechanism pins down which terms vanish and hence which objective is principled. When $U \rightarrow Y, U \not\rightarrow \mathbf{X}, Y \rightarrow \mathbf{X}$, the conditional covariate shift vanishes, $I(\mathbf{X}; E | Y) = 0$, so minimizing $I(\phi(\mathbf{X}); E | Y)$ is well-motivated. Symmetrically, when $U \not\rightarrow Y, U \rightarrow \mathbf{X}, \mathbf{X} \rightarrow Y$, the concept shift vanishes, $I(Y; E | \mathbf{X}) = 0$, motivating minimization of $I(Y; E | \phi(\mathbf{X}))$. In the confounded case $U \rightarrow Y$ and $U \rightarrow \mathbf{X}$, no such term vanishes a priori, the terms of Equation (1) interact nontrivially, and the risk-optimal strategy is not determined by the graph alone. The next section resolves this case.

4.2 Predictive information under hidden confounding shift

To characterize the effect of a confounder U acting jointly on $\mathbf{X}$ and Y , we analyze the *two exhaustive orientations* of the $\mathbf{X}$ – Y edge: (i) $\mathbf{X} \rightarrow Y$ and (ii) $Y \rightarrow \mathbf{X}$. Under $\mathbf{X} \rightarrow Y$, the inequalities in (2) follow from d-separation: conditioning on Y opens the path $\phi(\mathbf{X}) \leftarrow \mathbf{X} \rightarrow Y \leftarrow U \leftarrow E$ at the collider Y , admitting additional information flow from $\phi(\mathbf{X})$ to E through Y , while conditioning on $\phi(\mathbf{X})$ partially blocks the flow from Y to E at $\mathbf{X}$ to the extent that $\phi(\mathbf{X})$ retains information about $\mathbf{X}$.

$$(i) \quad \overbrace{I(\phi(\mathbf{X}); E|Y)}^{\text{Variation}} \geq \overbrace{I(\phi(\mathbf{X}); E)}^{\text{Feature shift}} \quad (ii) \quad \overbrace{I(Y; E)}^{\text{Label shift}} \geq \overbrace{I(Y; E|\phi(\mathbf{X}))}^{\text{Concept shift}} \quad (2)$$

Symmetrically, under $Y \rightarrow \mathbf{X}$, the inequalities in (3) follow: conditioning on Y blocks the flow from $\phi(\mathbf{X})$ to E at Y , while conditioning on $\phi(\mathbf{X})$ —a child of the collider $\mathbf{X}$ —opens the path $Y \rightarrow \mathbf{X} \leftarrow U \leftarrow E$ from Y to E , since conditioning on a descendant of a collider activates the collider path [58] (cf. the d-separation rules in § 4.1).

$$(i) \quad \overbrace{I(\phi(\mathbf{X}); E)}^{\text{Feature shift}} \geq \overbrace{I(\phi(\mathbf{X}); E|Y)}^{\text{Variation}} \quad (ii) \quad \overbrace{I(Y; E|\phi(\mathbf{X}))}^{\text{Concept shift}} \geq \overbrace{I(Y; E)}^{\text{Label shift}} \quad (3)$$

Substituting the inequalities in (2) and (3) into Equation (1) collapses the shift-related terms and yields the following refinement.

293 **Proposition 4.2.** Let $\mathbf{X}$ be a covariate vector, Y a label, E the environment variable, ϕ a representa-
 294 tion map, $\hat{Y}$ the prediction, and U an unobserved confounder acting on both $\mathbf{X}$ and Y . If either (i)
 295 $\mathbf{X} \rightarrow Y$ or (ii) $Y \rightarrow \mathbf{X}$, then the predictive information $I(Y; \hat{Y})$ reduces to

$$I(Y; \hat{Y}) = \overbrace{I(\phi(\mathbf{X}); Y|E)}^{\text{Cond. informativeness}} - \overbrace{I(\phi(\mathbf{X}); Y|\hat{Y})}^{\text{Residual}} \quad (4)$$

296 Proposition 4.2 treats the orientations $\mathbf{X} \rightarrow Y$ and $Y \rightarrow \mathbf{X}$ separately, but the former predominates
 297 in real-world tabular prediction: 11 of the 16 datasets in recent benchmarks [57, 50, 24] exhibit the
 298 structure $\mathbf{X} \rightarrow Y$, and in the remaining 5 the number of covariates caused by Y is far smaller than
 299 the number causing Y . We state the operational consequence of Proposition 4.2 next.

Relative to Equation (1), the refined decomposition in (4)—valid for the two confounded configurations (i) $U \rightarrow \mathbf{X}, U \rightarrow Y, \mathbf{X} \rightarrow Y$ and (ii) $U \rightarrow \mathbf{X}, U \rightarrow Y, Y \rightarrow \mathbf{X}$ —annihilates the *variation, label-shift, feature-shift, and concept-shift* terms. Consequently, under hidden confounding shift with either edge orientation, maximizing the predictive information is exactly maximizing the difference *conditional informativeness* – *residual*.

300
 301 **Maximizing conditional informativeness:** Prashant et al. [62] propose a mixture-of-experts (MoE)
 302 estimator for OOD generalization under hidden confounding shift in which each expert is indexed
 303 by a value of the hidden confounder and minimizes risk within the corresponding environment.
 304 Maximizing per-environment predictive content is exactly maximizing the conditional informativeness
 305 $I(\phi(\mathbf{X}); Y|E)$, so Equation (4) furnishes a formal justification for MoE-type estimators. Beyond
 306 this, the decomposition exposes the restrictiveness of their operating assumptions and the need for
 307 estimators valid under more general shift: their guarantee requires overlapping confounder support,
 308 $\text{supp}(\mathbb{P}(U|\mathcal{E}_{te})) \subseteq \text{supp}(\mathbb{P}(U|\mathcal{E}_{tr}))$, together with discrete-valued proxies.

Figure 2 demonstrates that, even without confounder-support overlap and with a continuous-valued proxy—two conditions outside the guarantee of Prashant et al. [62]—a linear regression conditioned on environment-specific information recovers the correct cross-environment relationship between the observed covariates and the target. This establishes, by construction, the need for OOD generalization estimators that remain valid under limited confounder support and continuous confounders.

309
 310 **Minimizing the residual:** The residual $I(\phi(\mathbf{X}); Y | \hat{Y})$ is the label-relevant information in $\phi(\mathbf{X})$
 311 not transmitted by $\hat{Y}$, and it admits the entropy expansion $I(\phi(\mathbf{X}); Y | \hat{Y}) = H(Y | \hat{Y}) - H(Y | \phi(\mathbf{X}), \hat{Y})$. The first term is tied to the expected cross-entropy loss through the proper-scoring-rule
 312 identity $\mathbb{E}_{Y, \hat{Y}}[\ell_{\text{CE}}(Y, \hat{Y})] = H(Y | \hat{Y}) + \text{calibration error}$ [12, 10, 29], in which $H(Y | \hat{Y})$ is the
 313 *refinement error* measuring class separability. Minimizing the residual therefore decomposes into two
 314 complementary routes. The first drives $I(\phi(\mathbf{X}); Y | \hat{Y})$ to zero directly by enforcing the conditional
 315 independence $Y \perp\!\!\!\perp \phi(\mathbf{X}) | \hat{Y}$ —that is, requiring $\hat{Y}$ to be a sufficient statistic of $\phi(\mathbf{X})$ for Y . The
 316 second minimizes the refinement error $H(Y | \hat{Y})$, e.g. via training schedules that reduce refinement
 317 error before addressing calibration error [10].
 318

319 4.3 Impact of additional informative variables

320 § 4.2 characterizes the risk-optimal strategy when the confounder is entirely unobserved. In many
 321 applications, however, one has access to proxies for the hidden confounder [4, 85, 62] that partially
 322 substitute for it. We now quantify the effect on predictive information of appending such non-causal
 323 but informative covariates to the conditioning set.

324 **Definition 4.3** (Informative Covariates). A set of covariates $\mathbf{X}_I$ that are not causally related to Y i.e.,
 325 neither ancestors nor descendants of Y are in $\mathbf{X}_I$, are said to be informative to Y if $\mathbf{X}_I$ and Y are
 326 not independent of each other given other causally related covariates $\mathbf{X}$ and E i.e., $Y \not\perp\!\!\!\perp \mathbf{X}_I | \mathbf{X}, E$.

327 Because $U \rightarrow Y$, informativeness for U and informativeness for Y coincide. Equation (1) implies
 328 that maximizing $I(Y; \hat{Y})$ requires minimizing the concept shift $I(Y; E | \phi(\mathbf{X}))$. Liu et al. [50] report

Table 2: Estimated distribution-shift magnitudes (mutual-information terms, in nats). Each entry is the mean $\pm$ standard deviation over 10 random subsets of 40,000 samples. A one-sample t -test against a null mean of zero ($\alpha = 0.05$) rejects the null for every entry (p-value ≈ 0), so all four shifts are significantly nonzero.

Dataset	Conditional Covariate Shift $I(\mathbf{X}; E Y)$	Label shift $I(Y; E)$	Covariate Shift $I(\mathbf{X}; E)$	Concept shift $I(Y; E \mathbf{X})$
Readmission	0.107 ± 0.002	0.068 ± 0.002	0.097 ± 0.002	2.032 ± 0.000
Food stamps	0.126 ± 0.004	0.030 ± 0.003	0.108 ± 0.001	2.118 ± 0.001
Income	0.168 ± 0.002	0.075 ± 0.003	0.147 ± 0.001	2.059 ± 0.002
Public coverage	0.231 ± 0.002	0.412 ± 0.006	0.222 ± 0.002	1.945 ± 0.001
Unemployment	0.117 ± 0.001	0.019 ± 0.002	0.114 ± 0.002	2.010 ± 0.003
Diabetes	0.032 ± 0.002	0.048 ± 0.001	0.022 ± 0.002	2.132 ± 0.001
Hypertension	0.090 ± 0.002	0.183 ± 0.003	0.037 ± 0.001	1.883 ± 0.004
ASSISTments	0.293 ± 0.002	0.260 ± 0.001	0.306 ± 0.002	0.367 ± 0.004

empirically that augmenting the covariate set with $\mathbf{X}_I$ so that $\mathbb{P}(Y|\mathbf{X}, \mathbf{X}_I)$ is more stable across environments—i.e. reducing concept shift—raises OOD accuracy. We establish the corresponding formal result: appending informative covariates moves each term of (1) monotonically in its predictive-information-increasing direction.

Proposition 4.3. *Let $\phi_1(\cdot), \phi_2(\cdot)$ be invertible maps and let $\mathbf{X}_I$ be informative variables satisfying $Y \not\perp \mathbf{X}_I \mid \mathbf{X}, E$. Then the following strict inequalities hold:*

$$\begin{aligned}
 (i) \quad & I(\phi_2(\{\mathbf{X} \cup \mathbf{X}_I\}); Y|E) > I(\phi_1(\mathbf{X}); Y|E) \quad (ii) \quad I(\phi_2(\{\mathbf{X} \cup \mathbf{X}_I\}); E) > I(\phi_1(\mathbf{X}); E) \\
 (iii) \quad & I(Y; E|\phi_2(\{\mathbf{X} \cup \mathbf{X}_I\})) < I(Y; E|\phi_1(\mathbf{X})) \quad (iv) \quad I(\phi_2(\{\mathbf{X} \cup \mathbf{X}_I\}); E \mid Y) > I(\phi_1(\mathbf{X}); E \mid Y)
 \end{aligned} \tag{5}$$

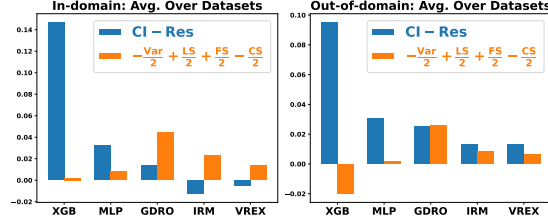
Proposition 4.3 establishes that appending informative covariates strictly increases conditional informativeness (5.i) and feature shift (5.ii) and strictly decreases concept shift (5.iii)—each change raising $I(Y; \hat{Y})$ per Equation (1). The same operation, however, strictly increases the variation term (5.iv), which enters with a negative coefficient and therefore calls for a dedicated mechanism to control it.

5 Experimental results

We conduct experiments on real-world and synthetic datasets to test three claims: (i) that hidden confounding shift is present and detectable in real-world data, (ii) that the terms of the decomposition in (1) govern OOD generalization under hidden confounding shift as the theory predicts, and (iii) that informative covariates increase generalization accuracy. The evaluation uses eight real-world tabular benchmarks—*Food stamps*, *Readmission*, *Income*, *Public coverage*, *Unemployment*, *Diabetes*, *Hypertension*, and *ASSISTments*—together with their domain splits, all taken from the *TableShift* benchmark [24]. Performance is reported as ID and OOD test accuracy. We evaluate two ERM-based estimators, XGBoost (XGB) [15] and a multi-layer perceptron (MLP); two domain-generalization estimators, IRM [6] and VREX [38]; and one robust-learning estimator, Group DRO (GDRO) [69]. Each reported value is the mean over 5 hyperparameter configurations sampled at random from the grids in Appendix B; the effect of the random seed is statistically negligible on these datasets [24]. The full protocol is specified in Appendix § B, and results under explicit hyperparameter tuning appear in Appendix C.

Detecting hidden confounding shift: As established in § 3, a hidden confounding shift is diagnosed by the simultaneous presence of shifts in $\mathbb{P}(\mathbf{X})$, $\mathbb{P}(\mathbf{X} \mid Y)$, $\mathbb{P}(Y)$, and $\mathbb{P}(Y \mid \mathbf{X})$, since no single-mechanism shift produces all four at once. Table 2 reports strictly positive estimates of covariate, conditional-covariate, label, and concept shift on every real-world dataset, satisfying this diagnostic and confirming the presence of hidden confounding shift. As a complementary qualitative check, we prompt GPT-4o [1] to enumerate candidate hidden confounders for several benchmarks [24, 57] (Appendix § E); for *food stamps*, for instance, it identifies unmeasured state-specific *economic policies* that plausibly act on both *household income* and *food stamp reciprocity*.

Conditional informativeness versus accuracy: Proposition 4.2 predicts that, under hidden confounding shift, the difference *conditional informativeness* – *residual* is the sole determinant of predictive information. We test this prediction on the six datasets in Table 2 that contain more than one training environment [57, 24]; the aggregate is in Figure 4 and the per-dataset breakdown in



Method	Avg. ID Test Acc.	Avg. OOD Test Acc.
XGB	85.22	77.00
MLP	83.64	73.29
GDRO	78.40	72.28
IRM	71.17	64.89
VREX	59.74	63.06

Figure 4: Across the eight datasets, the estimated difference *conditional informativeness* – *residual* (plots) is positively correlated with mean ID and OOD test accuracy (table, right), as predicted by Proposition 4.2.

Table 3: Sign consistency is uniformly high for concept shift (CS) across all estimators. By Proposition 4.2, however, the discriminating quantity is the sign consistency of conditional informativeness (CI), on which XGB attains the highest value. Abbreviations: Res residual, Var variation, FS feature shift; C causal, AC arguably causal, A all covariates.

	Sign Consistency Metric (↑)					ID-Test Accuracy (↑)			OOD-Test Accuracy (↑)		
Method	CI	Var	FS	CS	Res	C	AC	A	C	AC	A
XGB	0.89	0.61	0.36	0.89	0.17	81.24	84.84	85.22	69.92	76.72	77.00
MLP	0.75	0.64	0.28	0.89	0.33	80.51	81.93	83.64	69.70	73.84	73.29
GDRO	0.69	0.67	0.19	0.94	0.36	80.92	83.76	78.40	69.61	73.49	72.28
IRM	0.78	0.61	0.25	0.92	0.25	67.91	68.22	71.17	61.21	62.96	64.89
VREX	0.64	0.61	0.25	0.89	0.44	54.39	62.32	59.74	56.62	63.53	63.06

Appendix C. The estimated difference *conditional informativeness* – *residual* is positively correlated with both ID and OOD accuracy, and XGB—the estimator attaining the largest value of this difference—attains the highest accuracy. Consistent with Proposition 4.2, the per-dataset results (Appendix C) show the orange bars concentrated near zero, since under hidden confounding shift the aggregate $-\text{variation}/2 + \text{label shift}/2 + \text{feature shift}/2 - \text{concept shift}/2$ vanishes. The relative ID/OOD ordering of the estimators in Figure 1 reproduces the rankings reported in prior work [57, 24].

Informative covariates versus accuracy: We next quantify the contribution of informative covariates to OOD generalization. Following the partition of Nastl and Hardt [57], the available covariates form a nested chain of conditioning sets *causal covariates* $\subseteq$ *arguably causal covariates* $\subseteq$ *all covariates* (Appendix § B), and we track how the terms of Equation (1) move as the conditioning set is enlarged along this chain. We summarize the movement by a *sign-consistency metric* defined as follows. For a term with a positive coefficient in (1) (e.g. conditional informativeness), we record whether its estimate increases whenever accuracy increases; for a term with a negative coefficient (e.g. concept shift), whether its estimate decreases whenever accuracy increases. The metric is the empirical fraction of enlargements in which the term moves in its predictive-information-increasing direction, and thus measures how reliably that term drives accuracy gains; the formal definition is given in Appendix § B. Table 3 shows that conditional informativeness and concept shift both attain high sign consistency with accuracy gains. Reducing concept shift is necessary, but XGB additionally increases conditional informativeness beyond the other estimators, which accounts for its further accuracy advantage. Per-dataset results are reported in Appendix § C.

On real-world data one cannot verify that the *all-covariates* set exhausts the *relevant or sufficiently informative* variables. To control for this, we construct synthetic data with a fully specified causal structure— $U \rightarrow X, U \rightarrow Y, X \rightarrow Y$, and $U \rightarrow \mathbf{X}_I$, where $\mathbf{X}_I$ are informative covariates generated by the hidden confounder U —so that ground truth is known by construction. Appending $\mathbf{X}_I$ to the conditioning set increases conditional informativeness and feature shift and decreases concept shift (Figure 5), matching Proposition 4.3. The variation term stays identically zero, as expected for this data-generating process. The generative model and further results are given in Appendix § D.

6 Discussion, limitations, and future work

We decomposed the predictive information between predictions and outcomes into interpretable terms that isolate the factors bounding OOD generalization (Equation (1)). Our analysis shows that hidden confounding shift—manifested as coupled changes in the covariate and outcome distributions across environments—is generically unavoidable, and that no finite set of observed covariates or proxies fully removes the residual ambiguity it induces. Because confounding entangles covariate,

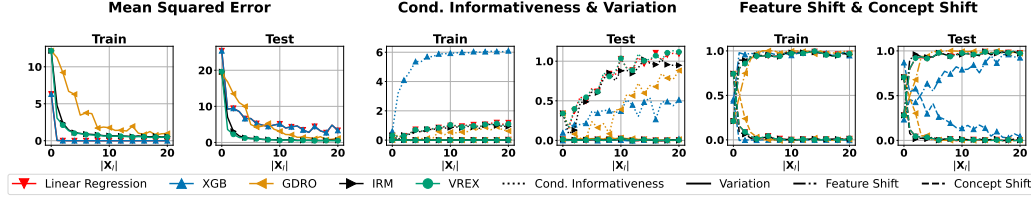


Figure 5: Appending proxy variables $\mathbf{X}_I$ of U that are informative for Y monotonically reduces MSE, increases conditional informativeness and feature shift, and reduces concept shift.

label, concept, and feature shift (inequalities (2) and (3)), objectives that target any one shift in isolation are provably inadequate. We proved instead that estimating environment-specific conditional relationships dominates enforcing a single invariant predictor (Equation (4)). By formalizing the inevitability of hidden confounding shift and the requirement to model it explicitly, this analysis supplies a basis for the next generation of OOD-robust estimators. The principal open problems are: (i) estimators with guarantees under limited confounder-support overlap; (ii) a formal characterization of the cost–accuracy trade-off in acquiring informative covariates; (iii) treatment of entangled shifts that does not rely on untestable proxy assumptions; and (iv) explicit modeling of the ambiguity contributed by unobserved confounders as a design principle for new OOD-robust paradigms.

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687 Appendix

688 A Proofs of propositions

689 **Proposition 4.1.** *Let $\mathbf{X}$ be a covariate vector and Y a label with causal structure $\mathbf{X} \leftrightarrow Y$ (a subset
690 of the covariates cause Y and the remainder are caused by Y), let E be the environment variable,
691 ϕ a representation map, and $\hat{Y}$ the prediction. Then the predictive information $I(Y; \hat{Y})$ admits the
692 exact decomposition*

$$I(Y; \hat{Y}) = \overbrace{I(\phi(\mathbf{X}); Y|E)}^{\text{Cond. informativ.}} - \frac{\overbrace{I(\phi(\mathbf{X}); E|Y)}^{\text{Variation}}}{2} + \frac{\overbrace{I(Y; E)}^{\text{Label shift}}}{2} + \frac{\overbrace{I(\phi(\mathbf{X}); E)}^{\text{Feature shift}}}{2} - \frac{\overbrace{I(Y; E|\phi(\mathbf{X}))}^{\text{Concept shift}}}{2} - \overbrace{I(\phi(\mathbf{X}); Y|\hat{Y})}^{\text{Residual}} \quad (1)$$

693 where the residual $I(\phi(\mathbf{X}); Y|\hat{Y})$ is the label-relevant information in $\phi(\mathbf{X})$ that the prediction $\hat{Y}$
694 fails to transmit. The identity holds verbatim with $\mathbf{X}$ substituted for $\phi(\mathbf{X})$.

695 *Proof.* From the causal graph shown in Figure 3 of the main paper, in the causal substructure:
696 $\hat{Y} \leftarrow \phi(\mathbf{X}) \leftarrow \mathbf{X} \leftrightarrow Y$, $\hat{Y}$ provides no additional information in predicting Y given either $\phi(\mathbf{X})$ or
697 $\mathbf{X}$. This is because, the path $\hat{Y} \leftarrow \phi(\mathbf{X}) \leftarrow \mathbf{X} \leftrightarrow Y$, between $\hat{Y}$ and Y is blocked by either $\{\phi(\mathbf{X})\}$
698 or $\{\mathbf{X}\}$. This implies $\hat{Y} \perp\!\!\!\perp Y|\phi(\mathbf{X})$ and $\hat{Y} \perp\!\!\!\perp Y|\mathbf{X}$. Now consider the expansion of $I(Y; \hat{Y})$.

$$\begin{aligned} I(Y; \hat{Y}) &= I(Y; \phi(\mathbf{X}), \hat{Y}) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Introduce } \phi(\mathbf{X})) \\ &= I(\phi(\mathbf{X}); Y) + \underbrace{I(Y; \hat{Y}|\phi(\mathbf{X}))}_0 - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Expand and use } \hat{Y} \perp\!\!\!\perp Y|\phi(\mathbf{X})) \\ &= I(\phi(\mathbf{X}), E; Y) - I(Y; E|\phi(\mathbf{X})) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Introduce } E) \\ &= I(Y; E) + I(\phi(\mathbf{X}); Y|E) - I(Y; E|\phi(\mathbf{X})) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Expand}) \end{aligned} \quad (6)$$

699 Similarly,

$$\begin{aligned} I(Y; \hat{Y}) &= I(Y; \phi(\mathbf{X}), \hat{Y}) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Introduce } \phi(\mathbf{X})) \\ &= I(\phi(\mathbf{X}); Y) + \underbrace{I(Y; \hat{Y}|\phi(\mathbf{X}))}_0 - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Expand and use } \hat{Y} \perp\!\!\!\perp Y|\phi(\mathbf{X})) \\ &= I(Y, E; \phi(\mathbf{X})) - I(\phi(\mathbf{X}); E|Y) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Introduce } E) \\ &= I(\phi(\mathbf{X}); E) + I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); E|Y) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Expand}) \end{aligned} \quad (7)$$

700 Summing Equations (6) and (7) and grouping the terms gives the desired expression:

$$\begin{aligned} I(Y; \hat{Y}) &= I(\phi(\mathbf{X}); Y|E) - \frac{I(\phi(\mathbf{X}); E|Y)}{2} + \frac{I(Y; E)}{2} \\ &\quad + \frac{I(\phi(\mathbf{X}); E)}{2} - \frac{I(Y; E|\phi(\mathbf{X}))}{2} - I(\phi(\mathbf{X}); Y|\hat{Y}). \end{aligned}$$

701 To get the similar decomposition with $\mathbf{X}$ instead of $\phi(\mathbf{X})$, use $\hat{Y} \perp\!\!\!\perp Y|\mathbf{X}$ and introduce $\mathbf{X}$ into the
702 expansion instead of $\phi(\mathbf{X})$. $\square$

703 **Proposition 4.2.** *Let $\mathbf{X}$ be a covariate vector, Y a label, E the environment variable, ϕ a representa-
704 tion map, $\hat{Y}$ the prediction, and U an unobserved confounder acting on both $\mathbf{X}$ and Y . If either (i)
705 $\mathbf{X} \rightarrow Y$ or (ii) $Y \rightarrow \mathbf{X}$, then the predictive information $I(Y; \hat{Y})$ reduces to*

$$I(Y; \hat{Y}) = \overbrace{I(\phi(\mathbf{X}); Y|E)}^{\text{Cond. informativeness}} - \overbrace{I(\phi(\mathbf{X}); Y|\hat{Y})}^{\text{Residual}} \quad (4)$$

706 *Proof.* Following the causal graph shown in Figure 3 of the main paper, we prove the proposition in
707 two separate cases: $\mathbf{X} \rightarrow Y$ and $Y \rightarrow \mathbf{X}$.

708 **Case 1: $\mathbf{X} \rightarrow Y$.** In this case, we have the following inequality because conditioning on Y opens the
 709 path $\phi(\mathbf{X}) \leftarrow \mathbf{X} \rightarrow Y \leftarrow U \leftarrow E$ between $\phi(\mathbf{X})$ and E at the collider node Y :

$$I(\phi(\mathbf{X}); E|Y) \geq I(\phi(\mathbf{X}); E). \quad (8)$$

710 Additionally, considering the causal substructure: $Y \leftarrow \mathbf{X} \leftarrow U \leftrightarrow E$, we have the following
 711 inequality because conditioning on $\phi(\mathbf{X})$ partially blocks the information flow from Y to E at the
 712 node $\mathbf{X}$ as long as $\phi(\mathbf{X})$ encodes some information about $\mathbf{X}$:

$$I(Y; E) \geq I(Y; E|\phi(\mathbf{X})). \quad (9)$$

713 Now consider the following expansion from Equation (7) in the proof of Proposition 4.1.

$$\begin{aligned} I(Y; \hat{Y}) &= I(\phi(\mathbf{X}); E) + I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); E|Y) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Equation (7)}) \\ &\leq I(\phi(\mathbf{X}); Y|E) + I(\phi(\mathbf{X}); E|Y) - I(\phi(\mathbf{X}); E|Y) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Using (8)}) \\ &= I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); Y|\hat{Y}) \end{aligned}$$

714 Similarly, consider the following expansion from Equation (6) from the proof of Proposition 4.1:

$$\begin{aligned} I(Y; \hat{Y}) &= I(Y; E) + I(\phi(\mathbf{X}); Y|E) - I(Y; E|\phi(\mathbf{X})) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Equation (6)}) \\ &\geq I(Y; E) + I(\phi(\mathbf{X}); Y|E) - I(Y; E) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Using (9)}) \\ &= I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); Y|\hat{Y}) \end{aligned}$$

715 That is, we have $I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); Y|\hat{Y}) \leq I(Y; \hat{Y}) \leq I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); Y|\hat{Y})$.

716 **Case 2: $Y \rightarrow \mathbf{X}$.** In this scenario, we have the following inequality because conditioning on $\phi(\mathbf{X})$
 717 opens the path $E \rightarrow U \rightarrow \mathbf{X} \leftarrow Y$ between Y and E at the collider node $\mathbf{X}$ because $\phi(\mathbf{X})$ is a child
 718 of $\mathbf{X}$ [58].

$$I(Y; E|\phi(\mathbf{X})) \geq I(Y; E) \quad (10)$$

719 Additionally, we have the following inequality because conditioning on Y blocks the path $\phi(\mathbf{X}) \leftarrow$
 720 $\mathbf{X} \leftarrow Y \leftarrow U \leftrightarrow E$ at Y and hence less information flow between $\phi(\mathbf{X})$ and E .

$$I(\phi(\mathbf{X}); E) \geq I(\phi(\mathbf{X}); E|Y) \quad (11)$$

721 Now consider the following expansion from Equation (6) from the proof of Proposition 4.1.

$$\begin{aligned} I(Y; \hat{Y}) &= I(Y; E) + I(\phi(\mathbf{X}); Y|E) - I(Y; E|\phi(\mathbf{X})) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Equation (6)}) \\ &\leq I(\phi(\mathbf{X}); Y|E) + I(Y; E|\phi(\mathbf{X})) - I(Y; E|\phi(\mathbf{X})) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Using (10)}) \\ &= I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); Y|\hat{Y}) \end{aligned}$$

722 Similarly, consider the following expansion from Equation (7) from the proof of Proposition 4.1.

$$\begin{aligned} I(Y; \hat{Y}) &= I(\phi(\mathbf{X}); E) + I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); E|Y) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Equation (7)}) \\ &\geq I(\phi(\mathbf{X}); Y|E) + I(\phi(\mathbf{X}); E) - I(\phi(\mathbf{X}); E) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Using (11)}) \\ &= I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); Y|\hat{Y}) \end{aligned}$$

723 That is, we have $I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); Y|\hat{Y}) \leq I(Y; \hat{Y}) \leq I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); Y|\hat{Y})$.

724 Since the upper bound and lower bounds are the same in both cases, we have the desired result
 725 $I(Y; \hat{Y}) = I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); Y|\hat{Y})$. $\square$

726 **Proposition 4.3.** Let $\phi_1(\cdot), \phi_2(\cdot)$ be invertible maps and let $\mathbf{X}_I$ be informative variables satisfying
 727 $Y \not\perp\!\!\!\perp \mathbf{X}_I \mid \mathbf{X}, E$. Then the following strict inequalities hold:

$$\begin{aligned} (i) & I(\phi_2(\{\mathbf{X} \cup \mathbf{X}_I\}); Y|E) > I(\phi_1(\mathbf{X}); Y|E) \quad (ii) \quad I(\phi_2(\{\mathbf{X} \cup \mathbf{X}_I\}); E) > I(\phi_1(\mathbf{X}); E) \\ (iii) & I(Y; E|\phi_2(\{\mathbf{X} \cup \mathbf{X}_I\})) < I(Y; E|\phi_1(\mathbf{X})) \quad (iv) \quad I(\phi_2(\{\mathbf{X} \cup \mathbf{X}_I\}); E \mid Y) > I(\phi_1(\mathbf{X}); E \mid Y) \end{aligned} \quad (5)$$

728 *Proof.* (i) By the chain rule of mutual information, we have: $I(\mathbf{X}, \mathbf{X}_I; Y|E) = I(\mathbf{X}; Y|E) +$
729 $I(\mathbf{X}_I; Y|E, \mathbf{X})$. We have $I(\mathbf{X}_I; Y|E, \mathbf{X}) > 0$ because $Y \not\perp\!\!\!\perp \mathbf{X}_I \mid \mathbf{X}, E$. Hence, we have
730 $I(\mathbf{X}, \mathbf{X}_I; Y|E) > I(\mathbf{X}; Y|E)$. Since $\phi_1(\cdot), \phi_2(\cdot)$ are invertible functions and mutual information is
731 invariant to invertible transformations, we have $I(\phi_2(\{\mathbf{X} \cup \mathbf{X}_I\}); Y|E) > I(\phi_1(\mathbf{X}); Y|E)$.

732 (ii) We consider the causal graph shown in Figure 3. From the chain rule of mutual information,
733 we have: $I(\mathbf{X}, \mathbf{X}_I; E) = I(\mathbf{X}; E) + I(\mathbf{X}_I; E|\mathbf{X})$. Since $E \leftrightarrow U \rightarrow Y$, any covariate that is
734 informative to Y is also informative to U . Since $E \leftrightarrow U$, we have $I(\mathbf{X}_I; E|\mathbf{X}) > 0$, and hence
735 $I(\mathbf{X}, \mathbf{X}_I; E) > I(\mathbf{X}; E)$. Since $\phi_1(\cdot), \phi_2(\cdot)$ are invertible functions and mutual information is
736 invariant to invertible transformations, we have $I(\phi_2(\{\mathbf{X} \cup \mathbf{X}_I\}); E) > I(\phi_1(\mathbf{X}); E)$.

737 (iii) Consider $I(Y; E, \mathbf{X}_I | \mathbf{X})$. It can be expressed in two ways:

$$I(Y; E, \mathbf{X}_I | \mathbf{X}) = I(Y; E | \mathbf{X}) + I(Y; \mathbf{X}_I | E, \mathbf{X}) = I(Y; \mathbf{X}_I | \mathbf{X}) + I(Y; E | \mathbf{X}_I, \mathbf{X})$$

738 Equating and rearranging the terms gives the following.

$$I(Y; E | \mathbf{X}) - I(Y; E | \mathbf{X}_I, \mathbf{X}) = I(Y; \mathbf{X}_I | \mathbf{X}) - I(Y; \mathbf{X}_I | E, \mathbf{X})$$

739 Since $Y \not\perp\!\!\!\perp \mathbf{X}_I \mid E, \mathbf{X}$, we have $I(Y; \mathbf{X}_I | \mathbf{X}) > I(Y; \mathbf{X}_I | E, \mathbf{X})$ because conditioning usually
740 reduces mutual information unless the additional conditioning variable opens any collider paths. Here
741 E is neither a collider nor a descendant of any collider. Thus, $I(Y; E | \mathbf{X}) > I(Y; E | \mathbf{X}_I, \mathbf{X})$. Since
742 $\phi_1(\cdot), \phi_2(\cdot)$ are invertible functions and mutual information is invariant to invertible transformations,
743 we have $I(Y; E | \phi_1(\mathbf{X})) > I(Y; E | \phi_2(\{\mathbf{X}_I \cup \mathbf{X}\}))$.

744 (iv) By the chain rule of mutual information, we have: $I(\mathbf{X}, \mathbf{X}_I; E | Y) = I(\mathbf{X}; E | Y) +$
745 $I(\mathbf{X}_I; E|\mathbf{X}, Y)$. Since $E \leftrightarrow U \rightarrow Y$, any covariate that is informative to Y is also informative to
746 U . Since $E \leftrightarrow U$, we have $I(\mathbf{X}_I; E|\mathbf{X}, Y) > 0$, and hence $I(\mathbf{X}, \mathbf{X}_I; E | Y) > I(\mathbf{X}; E | Y)$. Since
747 $\phi_1(\cdot), \phi_2(\cdot)$ are invertible functions and mutual information is invariant to invertible transformations,
748 we have $I(\phi_2(\{\mathbf{X} \cup \mathbf{X}_I\}); E | Y) > I(\phi_1(\mathbf{X}); E | Y)$. $\square$

749 B Experimental setup

750 This section specifies the experimental protocol. The random seed has statistically negligible effect in
751 most configurations [24]. For the main-paper results, each entry is the mean and standard deviation
752 over five hyperparameter configurations drawn uniformly at random from the grids in Table B1, per
753 dataset–estimator pair. Mutual-information terms are estimated with the nonparametric k -nearest-
754 neighbor entropy estimator [37, 77, 78]. Following Gardner et al. [24], every mutual-information
755 quantity is computed twice: on the ID test data (from the training domains) and on the OOD test data
756 (from the held-out domains).

757 For comparability across datasets we draw
758 20,000 samples at random from each of the ID
759 and OOD test sets, subsampling fewer where
760 a dataset is smaller. The estimated mutual-
761 information terms are largely insensitive to this
762 sample size: Figure B1 shows that the terms of
763 Equation (1), computed on the XGB representa-
764 tions for the *income* dataset, remain stable as the
765 sample size is varied. The representation $\phi(\mathbf{X})$
766 is taken from the layer immediately preceding
767 the classification head; for XGBoost, we use
768 the concatenation of the output margins and the
769 SHAP values. All experiments were run on a single NVIDIA RTX A6000 GPU.

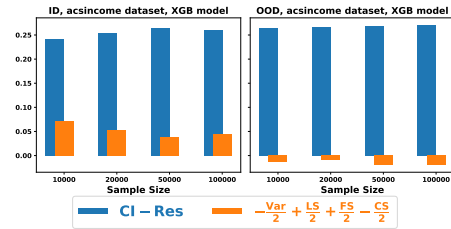


Figure B1: The estimated predictive-information terms are approximately invariant to sample size on the *income* dataset.

770 For the experiments quantifying the effect of informative covariates on predictive informa-
771 tion, we adopt the covariate taxonomy of Nastl and Hardt [57], comprising *causal*, *arguably*
772 *causal*, and *all* covariates. *Causal* covariates have an established directed effect on the tar-
773 get with well-supported orientation and implausible reverse causation; restricting to them, how-
774 ever, risks omitting true parents that are unknown. *Arguably causal* covariates satisfy at least
775 one of three criteria: (1) known causal, (2) plausible but possibly bidirectional influence, or
776 (3) likely-but-unconfirmed causal effect; they approximate the true parent set under available
777 knowledge, subject to residual confounding. Certain datasets further contain *anti-causal* co-
778 variates, which the target influences but which do not influence the target. The *all*-covariates

Table B1: Hyperparameters and their candidate search values per estimator.

Model	Hyperparameter	Search values
XGBoost	Learning rate	[0.1, 0.2, 0.3]
	Maximum tree depth	[4, 5, 6]
	Minimum child weight	[0.1, 1, 10]
	Gamma (min. loss reduction)	[0.0001, 0.001, 0.01]
	Subsample ratio	[0.5, 0.6, 0.7]
	Column subsample per tree	[0.5, 0.6, 0.7]
	L1 regularization (α)	[0.0001, 0.001, 0.01]
	L2 regularization (λ)	[0.0001, 0.001, 0.01]
MLP	Number of layers	[2, 3, 4]
	Hidden layer size	[256, 512, 1024]
	Dropout rate	[0.0, 0.1, 0.2]
	Learning rate	[0.01, 0.02, 0.05]
	Weight decay (L2)	[0.0001, 0.001, 0.01]
	Batch size	[4096]
	Number of epochs	[1, 2, 3]
GroupDRO	Number of layers	[2, 3, 4]
	Hidden layer size	[256, 512, 1024]
	Group-weights step size	[0.01, 0.05, 0.1]
	Dropout rate	[0.0, 0.1, 0.2]
	Learning rate	[0.01, 0.02, 0.05]
	Weight decay (L2)	[0.0001, 0.001, 0.01]
	Batch size	[4096]
IRM	Number of layers	[2, 3, 4]
	Hidden layer size	[256, 512, 1024]
	Dropout rate	[0.0, 0.1, 0.2]
	IRM penalty weight (λ)	[0.01, 0.05, 0.1]
	IRM penalty anneal iterations	[1, 2, 3]
	Learning rate	[0.01, 0.02, 0.05]
	Weight decay (L2)	[0.0001, 0.001, 0.01]
	Batch size	[4096]
VREX	Number of layers	[2, 3, 4]
	Hidden layer size	[256, 512, 1024]
	VREX penalty anneal iterations	[1, 2, 3]
	VREX penalty weight (λ)	[0.1, 10, 100]
	Dropout rate	[0.0, 0.1, 0.2]
	Learning rate	[0.01, 0.02, 0.05]
	Weight decay (L2)	[0.0001, 0.001, 0.01]
	Batch size	[4096]
VREX	Number of layers	[2, 3, 4]
	Hidden layer size	[256, 512, 1024]
	VREX penalty anneal iterations	[1, 2, 3]
	VREX penalty weight (λ)	[0.1, 10, 100]
	Dropout rate	[0.0, 0.1, 0.2]
	Learning rate	[0.01, 0.02, 0.05]
	Weight decay (L2)	[0.0001, 0.001, 0.01]
	Batch size	[4096]
VREX	Number of layers	[2, 3, 4]
	Hidden layer size	[256, 512, 1024]
	VREX penalty anneal iterations	[1, 2, 3]
	VREX penalty weight (λ)	[0.1, 10, 100]
	Dropout rate	[0.0, 0.1, 0.2]
	Learning rate	[0.01, 0.02, 0.05]
	Weight decay (L2)	[0.0001, 0.001, 0.01]
	Batch size	[4096]

779 set comprises every observed variable irrespective of causal role. The sign-consistency met-
780 ric of the main-paper § 5 is defined as follows. For each signed term $(m, \sigma_m) \in M$, where
781 $M = \{(\text{conditional informativeness}, +1), (\text{variation}, -1), (\text{label shift}, +1), (\text{feature shift}, +1),$
782 $(\text{concept shift}, -1), \text{ and } (\text{residual}, -1)\}$, for a specific dataset, we compute:

$$C_m = \frac{1}{2|P|} \sum_{(s,t) \in P} [\mathbb{I}(\sigma_m \cdot (\mu_t^{\text{tr}}(m) - \mu_s^{\text{tr}}(m)) > 0) + \mathbb{I}(\sigma_m \cdot (\mu_t^{\text{te}}(m) - \mu_s^{\text{te}}(m)) > 0)]$$

783 Here P is the set of ordered conditioning-set pairs (s, t) with $t \supset s$ —for instance, s the causal set
784 and t the arguably-causal set—and the factor 2 in the normalizer $2|P|$ accounts for the two evaluation
785 splits (train and test). The quantity $\mu_s^{\text{tr/te}}(m)$ is the mean of term m under conditioning set s on

the training/test split, $\mathbb{I}(\cdot)$ the indicator, and $|P| = 3$ for the three nested sets $\{causal, arguably causal, all\}$. The reported metric averages C_m over datasets, $\bar{C}_m = \frac{1}{|D|} \sum_{d \in D} C_m^{(d)}$, where $|D|$ is the number of datasets.

C Additional results on real-world datasets

This section reports additional real-world results. Figures C2 and C3 give the per-dataset values of the two grouped quantities *conditional informativeness – residual* and *–variation / 2 + label shift / 2 + feature shift / 2 – concept shift / 2*, whose aggregate appears in Figure 4 (left). By Proposition 4.1, the sum of the blue and orange groups is positively correlated with overall accuracy. Under hidden confounding shift, the blue difference *conditional informativeness – residual* accounts for a larger share of the predictive information than the orange group *–variation / 2 + label shift / 2 + feature shift / 2 – concept shift / 2*, which motivates maximizing conditional informativeness as the primary objective under this regime.

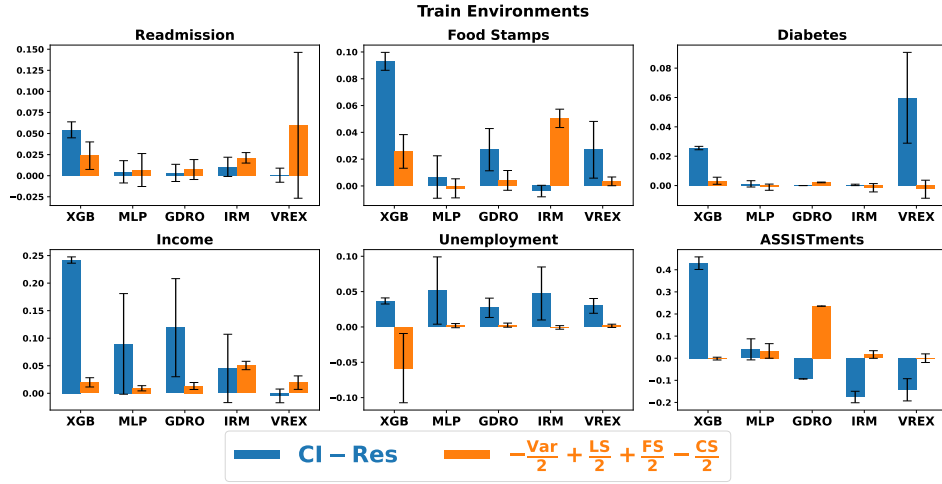


Figure C2: Per-dataset decomposition on ID test data.

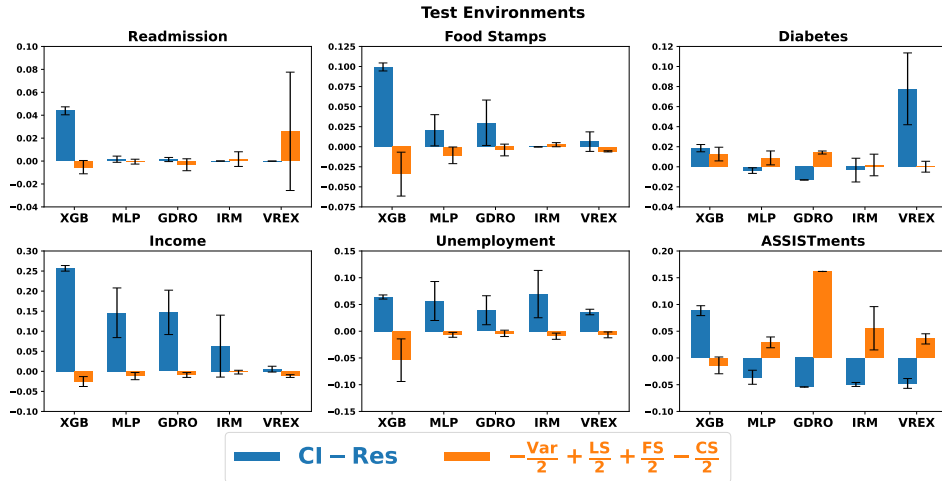


Figure C3: Per-dataset decomposition on OOD test data.

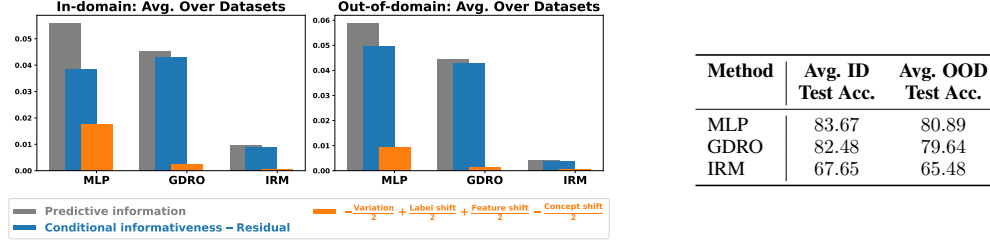


Figure C4: Under hyperparameter tuning, the estimated difference *conditional informativeness* – *residual* (plots) remains positively correlated with mean ID and OOD test accuracy over the five datasets (table, right). The *blue* difference *conditional informativeness* – *residual* contributes more to the predictive information than the *orange* group $-\text{variation} / 2 + \text{label shift} / 2 + \text{feature shift} / 2 - \text{concept shift} / 2$.

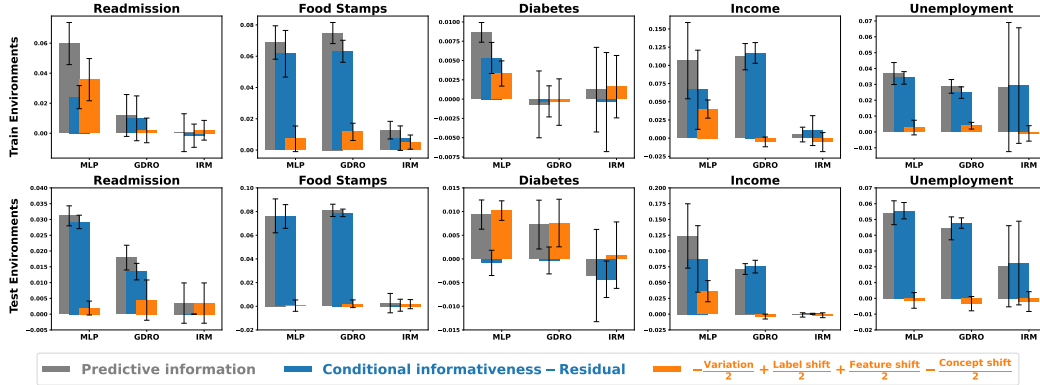


Figure C5: Per-dataset comparison of the predictive-information terms.

Table C2: Per-dataset ID and OOD test accuracy (mean $\pm$ standard deviation) by estimator.

Method	Readmission		Food stamps		Diabetes	
	ID test acc.	OOD test acc.	ID test acc.	OOD test acc.	ID test acc.	OOD test acc.
MLP	65.78 $\pm$ 0.14	61.75 $\pm$ 0.22	84.74 $\pm$ 0.03	82.00 $\pm$ 0.06	87.66 $\pm$ 0.02	83.22 $\pm$ 0.05
GDRO	60.47 $\pm$ 1.08	57.60 $\pm$ 0.42	84.47 $\pm$ 0.04	81.41 $\pm$ 0.09	87.47 $\pm$ 0.10	82.90 $\pm$ 0.10
IRM	50.28 $\pm$ 7.35	50.85 $\pm$ 1.74	80.91 $\pm$ 0.00	78.01 $\pm$ 0.00	42.54 $\pm$ 36.5	43.48 $\pm$ 31.9
	Income		Unemployment			
	ID test acc.	OOD test acc.	ID test acc.	OOD test acc.		
MLP	82.92 $\pm$ 0.05	81.46 $\pm$ 0.01	97.27 $\pm$ 0.02	96.04 $\pm$ 0.05		
GDRO	82.70 $\pm$ 0.05	80.16 $\pm$ 0.45	97.28 $\pm$ 0.01	96.10 $\pm$ 0.00		
IRM	67.91 $\pm$ 0.00	60.20 $\pm$ 0.00	96.61 $\pm$ 0.00	94.84 $\pm$ 0.00		

Figures C6–C15 give the term-wise decomposition of (1) for every estimator–dataset pair. The second column (*variation*) and third column (*feature shift*) are of comparable magnitude, so by Equation 1 their signed difference is near zero and contributes negligibly. Likewise, the fourth column (*label shift*) and fifth column (*concept shift*) are of comparable magnitude, and their signed difference is near zero. Consequently, in nearly every setting the difference *conditional informativeness* – *residual* is the dominant contributor to the predictive information.

Hyperparameter tuning: We additionally report results under explicit hyperparameter selection. We perform a grid search and select the configuration maximizing ID validation accuracy—not OOD validation accuracy—to avoid test-domain leakage. This yields 1500 runs (5 datasets $\times$ 3 estimators $\times$ 100 grid configurations). Means and standard deviations for the accuracies and mutual-information terms are computed over 5 random seeds per run, notwithstanding the negligible seed effect. Results appear in Figure C4. Consistent with Proposition 4.1, the sum of the *blue* and *orange* groups is positively correlated with accuracy, and the *blue* difference *conditional informativeness* – *residual*

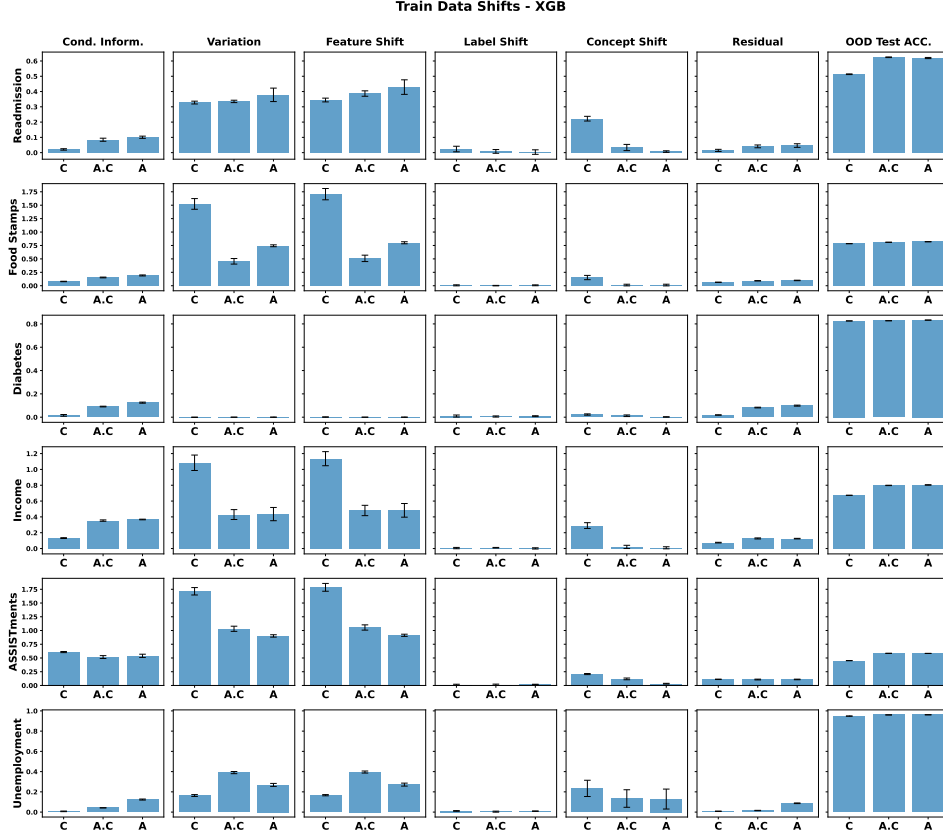


Figure C6: Term-wise decomposition of predictive information on ID test data for the XGBoost estimator, by conditioning set: (C) causal, (A.C) arguably causal, (A) all covariates.

811 contributes more to the predictive information than the **orange** group $-variation / 2 + label\ shift / 2 +$
812 $feature\ shift / 2 - concept\ shift / 2$. The per-dataset results in Figure C5 and Table C2 exhibit the
813 same trend.

814 **Limitations of the predictive-information decomposition:** The predictive information $I(Y; \hat{Y})$
815 measures statistical dependence between prediction and label but is not a monotone function of
816 accuracy. A binary classifier that deterministically inverts the label ($\hat{Y} = 1$ when $Y = 0$) attains the
817 maximal $I(Y; \hat{Y}) = 1$ bit through perfect anti-correlation while achieving 0% accuracy, precisely
818 because mutual information quantifies uncertainty reduction rather than correctness. The individual
819 terms admit analogous caveats: a large conditional informativeness $I(\phi(X); Y|E)$ can coexist with
820 large, harmful variation $I(\phi(X); E|Y)$, and feature shift $I(\phi(X); E)$ can inflate $I(Y; \hat{Y})$ via spurious
821 correlation. For estimators with non-degenerate accuracy—the regime of our experiments—the
822 decomposition nonetheless remains diagnostic: it attributes the predictive information to generalizable
823 versus environment-specific components and thereby directs targeted improvements while preserving
824 interpretability.

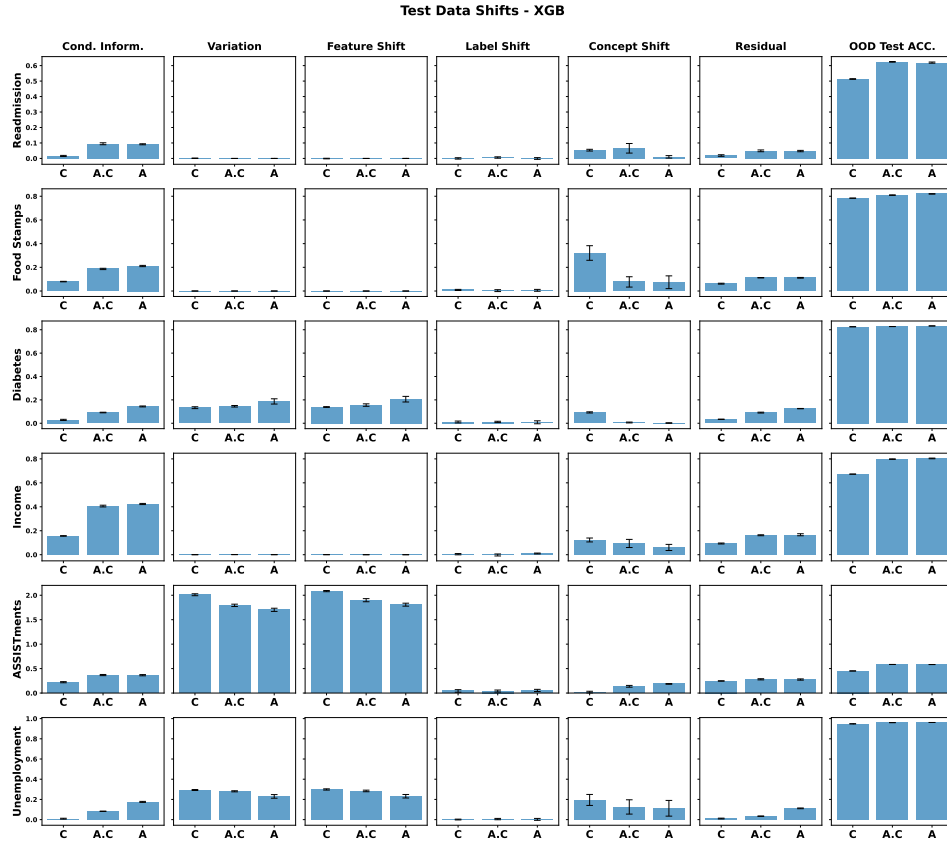


Figure C7: Term-wise decomposition of predictive information on OOD test data for the XGBoost estimator, by conditioning set: (C) causal, (A.C) arguably causal, (A) all covariates.

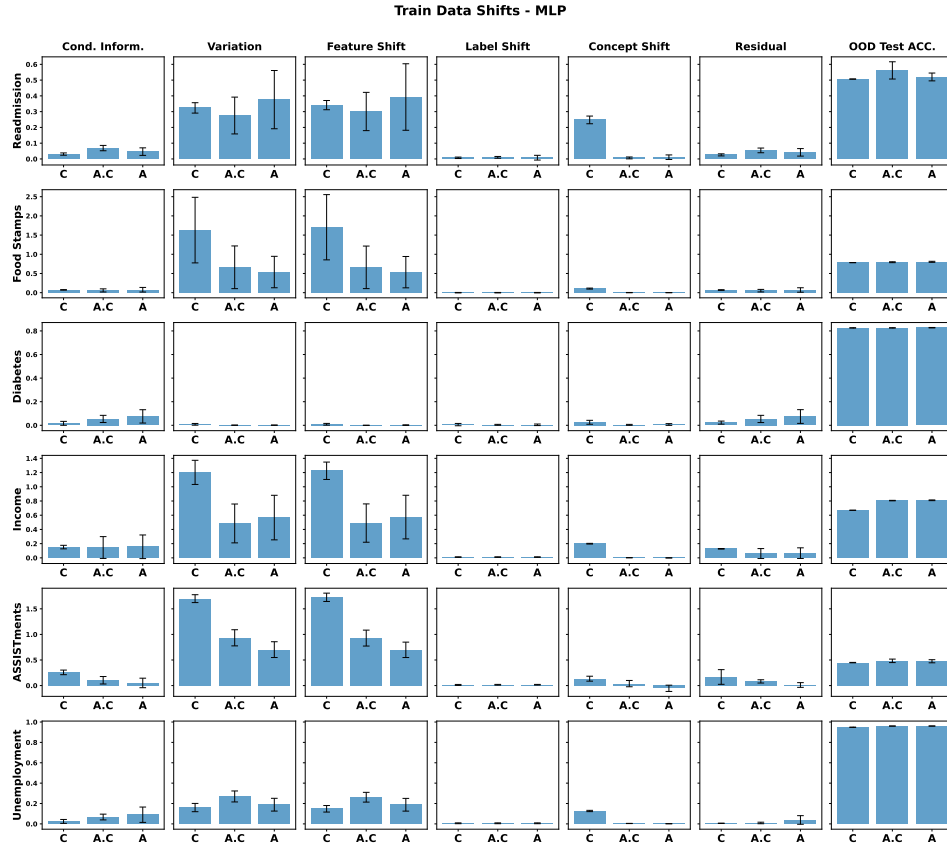


Figure C8: Term-wise decomposition of predictive information on ID test data for the MLP estimator, by conditioning set: (C) causal, (A.C) arguably causal, (A) all covariates.

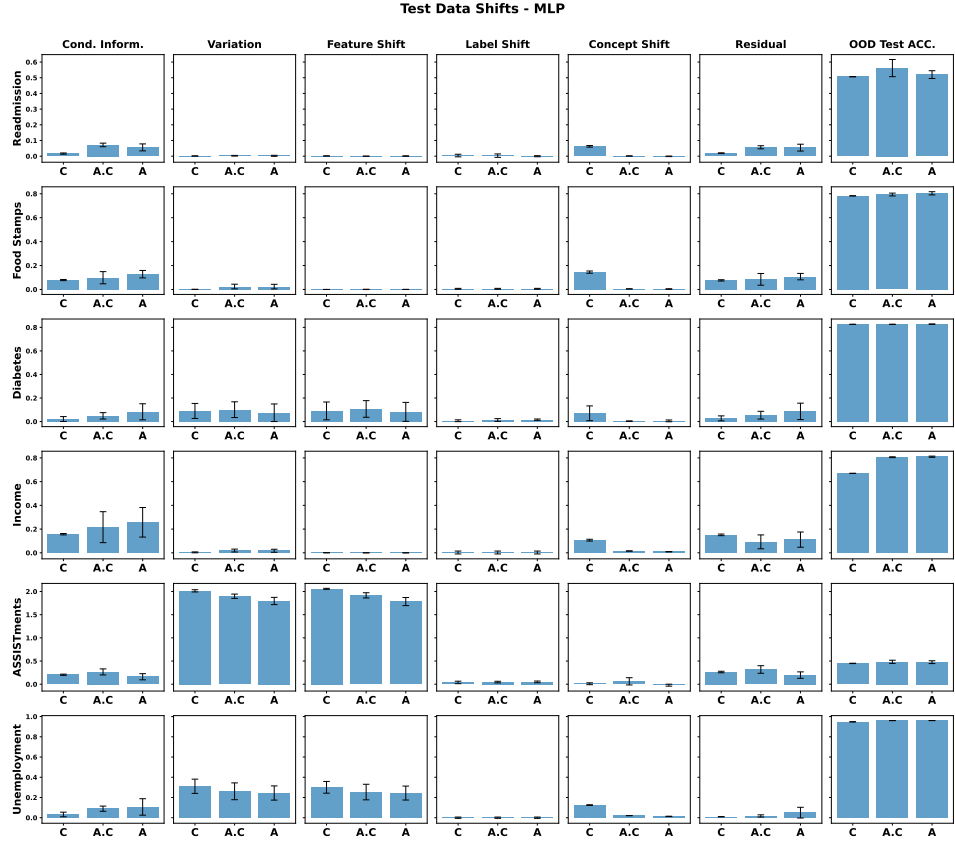


Figure C9: Term-wise decomposition of predictive information on OOD test data for the MLP estimator, by conditioning set: (C) causal, (A.C) arguably causal, (A) all covariates.

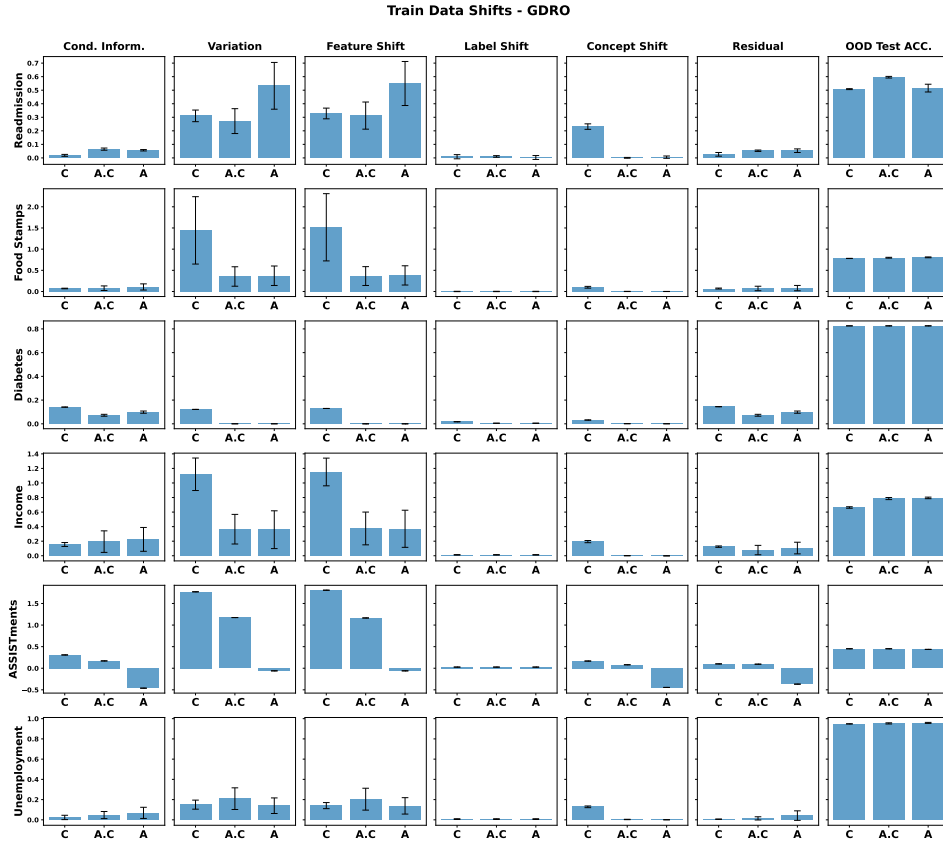


Figure C10: Term-wise decomposition of predictive information on ID test data for the GDRO estimator, by conditioning set: (C) causal, (A.C) arguably causal, (A) all covariates.

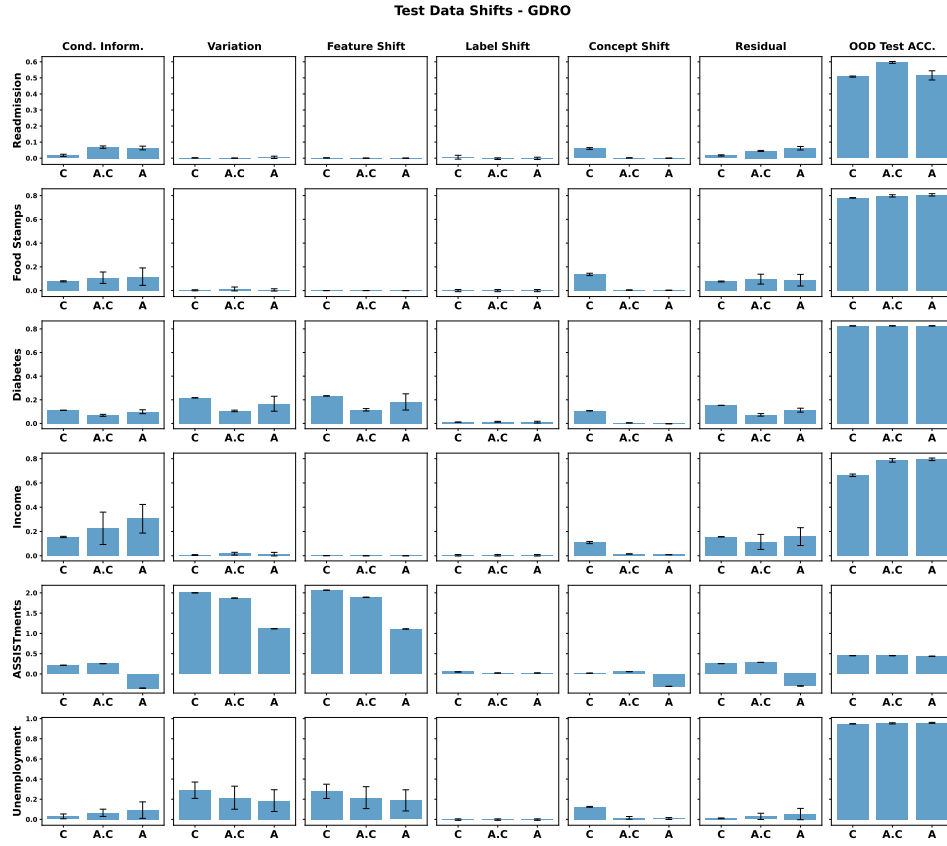


Figure C11: Term-wise decomposition of predictive information on OOD test data for the GDRO estimator, by conditioning set: (C) causal, (A.C) arguably causal, (A) all covariates.

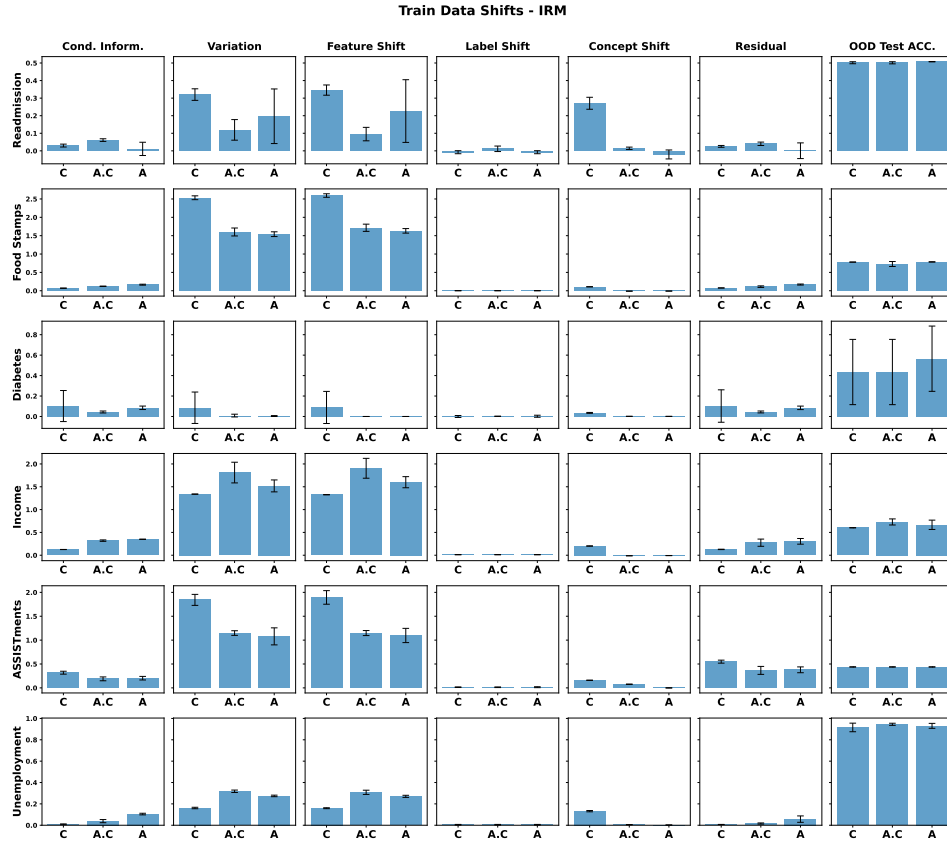


Figure C12: Term-wise decomposition of predictive information on ID test data for the IRM estimator, by conditioning set: (C) causal, (A.C) arguably causal, (A) all covariates.

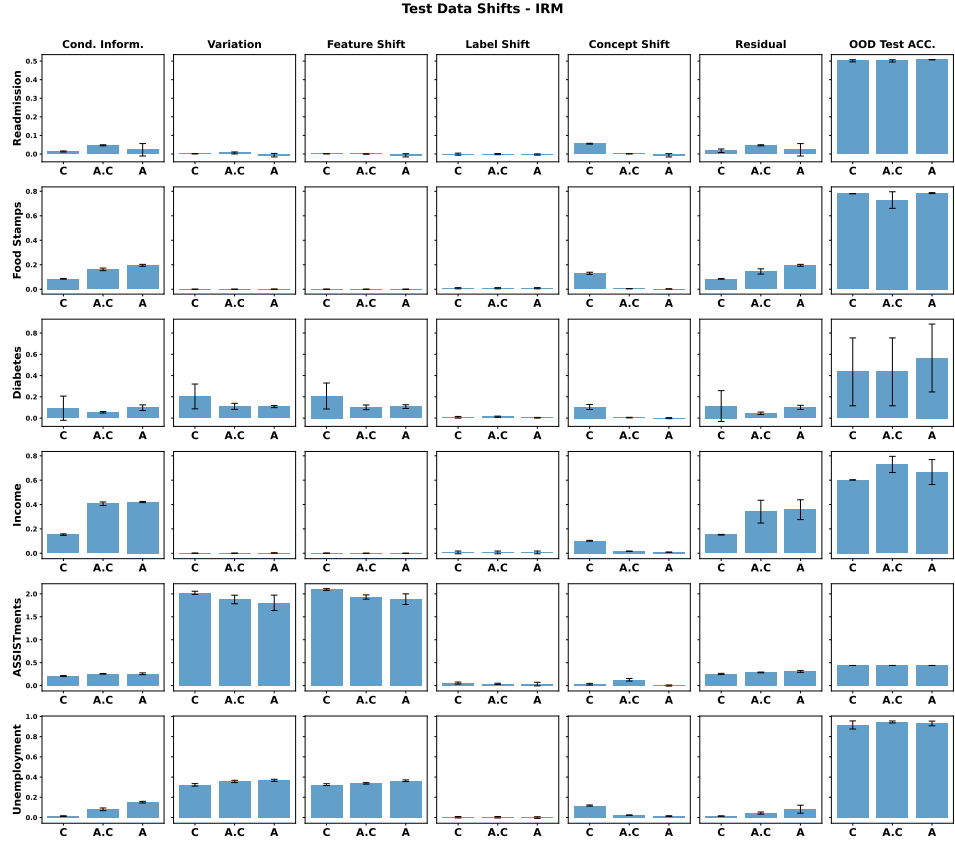


Figure C13: Term-wise decomposition of predictive information on OOD test data for the IRM estimator, by conditioning set: (C) causal, (A.C) arguably causal, (A) all covariates.

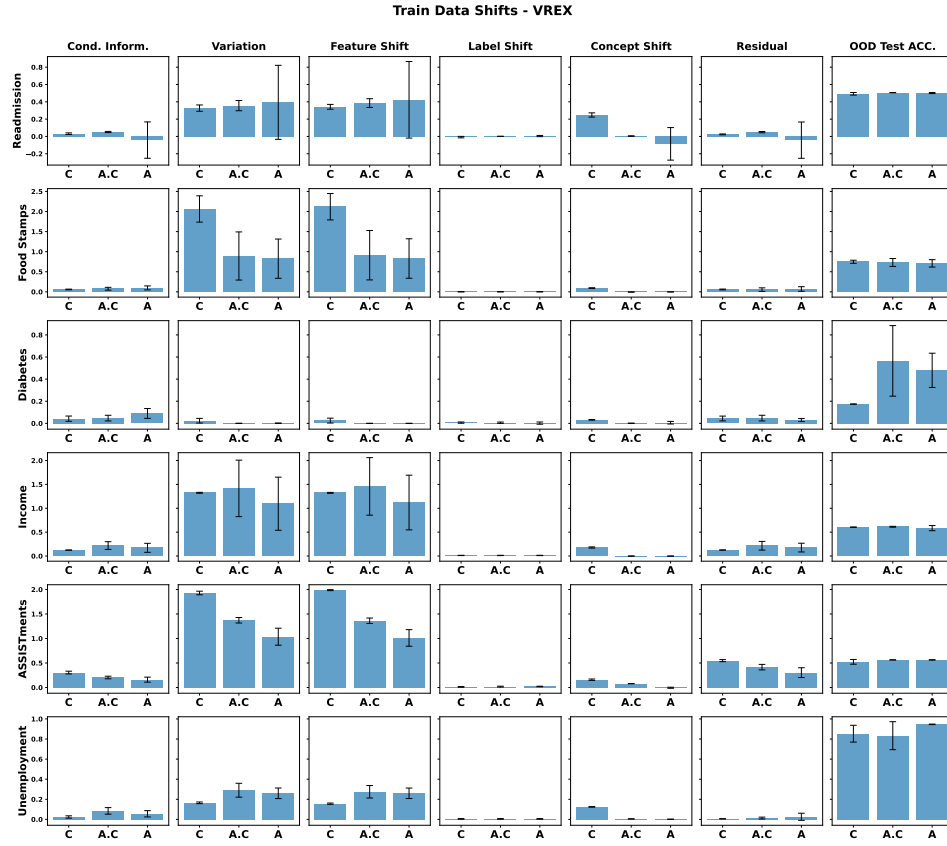


Figure C14: Term-wise decomposition of predictive information on ID test data for the VREX estimator, by conditioning set: (C) causal, (A.C) arguably causal, (A) all covariates.

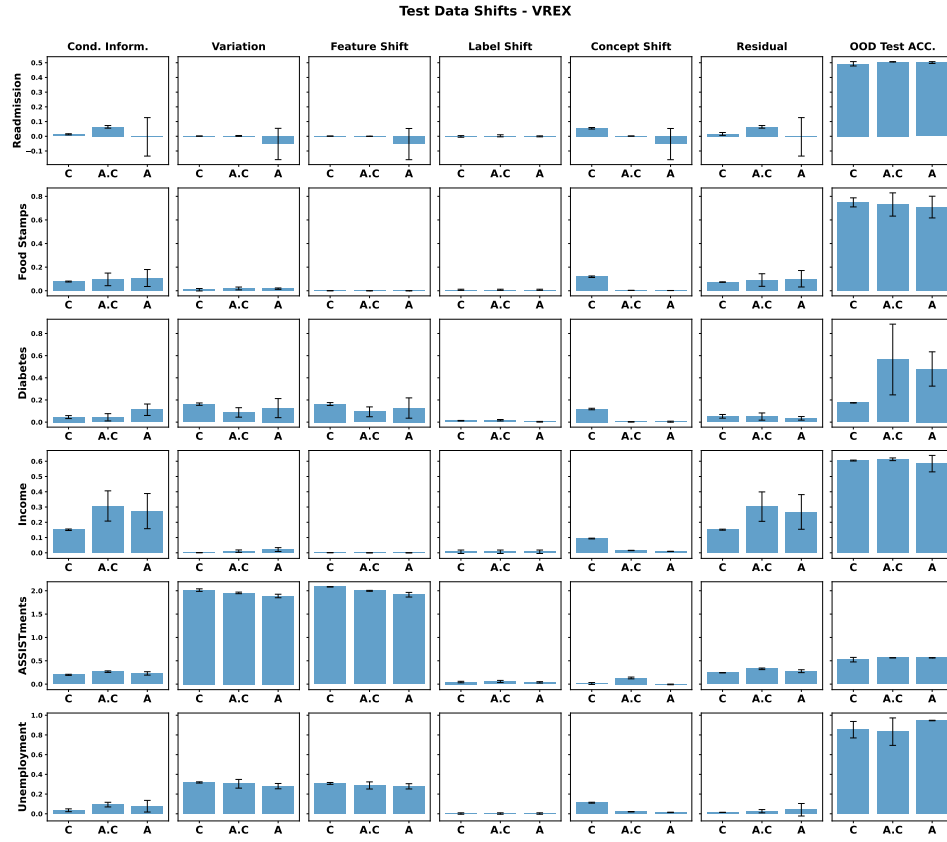


Figure C15: Term-wise decomposition of predictive information on OOD test data for the VREX estimator, by conditioning set: (C) causal, (A.C) arguably causal, (A) all covariates.

825 D Results on synthetic datasets

826 We run a suite of synthetic experiments to isolate three factors: (i) the choice of informativeness
827 criterion, (ii) the degree of confounder-support overlap across environments, and (iii) the ratio of
828 causal to anti-causal variables. Data are generated from the structural equations (12) under the causal
829 structure $U \rightarrow X, U \rightarrow Y, U \rightarrow \mathbf{X}_I$, and $X \rightarrow Y$, in which U is the unobserved confounder, X the
830 observed covariate, Y the target, and $\mathbf{X}_I$ the additional informative covariates.

$$\begin{aligned} U &\sim \mathcal{N}(\mu_u^e, \sigma_u) & X &\leftarrow f_X(U) + \mathcal{N}(\mu_x, \sigma_x) \\ X_i &\leftarrow f_i(U) + \mathcal{N}(\mu_i, \sigma_i); X_i \in \mathbf{X}_I & Y &\leftarrow f_Y(X, U) + \mathcal{N}(\mu_y, \sigma_y) \end{aligned} \quad (12)$$

831 We instantiate several functional forms for f_X , f_i , and f_Y , reporting the corresponding results below.
832 A distribution shift is induced by varying the environment-specific mean μ_u^e of the confounder
833 distribution, so that the separation between the training and test values of μ_u^e controls the magnitude
834 of the shift under study.

835 **Informativeness versus accuracy:** We first fix the linear structural equations $f_X = 0.3U$, $f_i = 0.1U$,
836 and $f_Y = X - 2U$, with training means $\mu_u^e \in \{-2, 2\}$ and test means $\mu_u^e \in \{0, 4\}$ —a configuration
837 with limited support overlap. For Figure 5 of the main paper we use a single confounder and
838 $|\mathbf{X}_I| = 20$ informative covariates. Increasing the number of appended informative covariates from
839 0 to 20 produces (i) a monotone decrease in mean squared error, (ii) an increase in conditional
840 informativeness, (iii) an increase in feature shift, and (iv) a decrease in concept shift—exactly the
841 movement predicted by Proposition 4.3. We probe two further informativeness criteria. First, holding
842 $|\mathbf{X}_I| = 1$ with one confounder, we sweep the covariate noise $\sigma_i \in \{0, 0.1, \dots, 2.0\}$: Figure D16
843 shows that decreasing σ_i from 2.0 to 0.0 yields the same four-way movement in MSE, conditional
844 informativeness, feature shift, and concept shift, with the variation term identically zero throughout.
845 Second, with 20 confounders and 20 matched informative covariates under $f_X = 0.3U$, $f_i = 0.2U$,
846 $f_Y = X - 2U$, Figure D17 reproduces the identical pattern as informative covariates are added.
847 Across all three criteria, the direction of every term matches the theory.

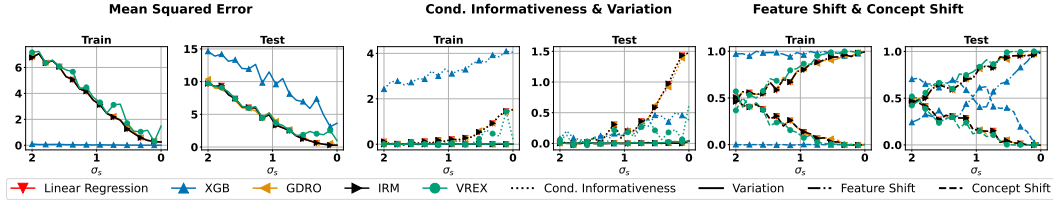


Figure D16: Appending a low-noise proxy variable reduces MSE, increases conditional informativeness and feature shift, and reduces concept shift.

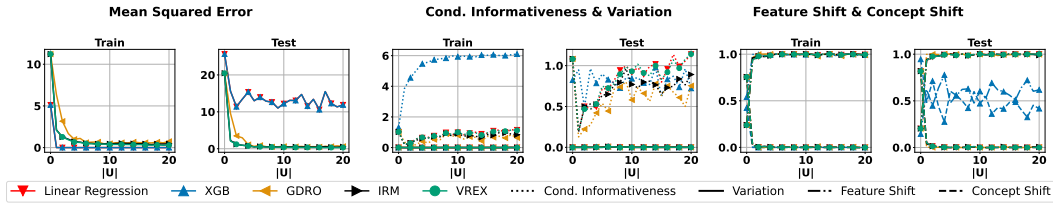


Figure D17: Appending proxy variables $\mathbf{X}_I$ for a set of hidden confounders $\mathbf{U}$ that are informative for Y reduces MSE, increases conditional informativeness and feature shift, and reduces concept shift.

848 **Confounder-support overlap versus accuracy:** The guarantee of Prashant et al. [62] requires the
849 test confounder support to be contained in the training support. Having shown recovery of the correct
850 relationship on synthetic linear data whenever the observed covariates carry sufficient confounder
851 information, we now extend the analysis to nonlinear data using MLP and XGB estimators under
852 both low- and high-overlap conditions (Figures D18–D21). Under low overlap (widely separated μ_e
853 between ID and OOD), the estimator conditioned on X alone fits a single global function that cannot

Structural Equations :

(Hidden Confounder) $U \sim \mathcal{N}(\mu_e, 0.5^2)$
 $\mu_e \in \{-2, 2\}$ for ID
 $\mu_e \in \{0, 4\}$ for OOD
(Input) $X = 0.75U + \mathcal{N}(0, 1.5^2)$
(Outcome) $Y = X^2 - 3U + \mathcal{N}(0, 0.5^2)$
(Informative) $X_i = 0.6U + \mathcal{N}(0, 0.3^2)$

▲ Train data
▼ Test data
★ Predictions on train data
✱ Predictions on test data

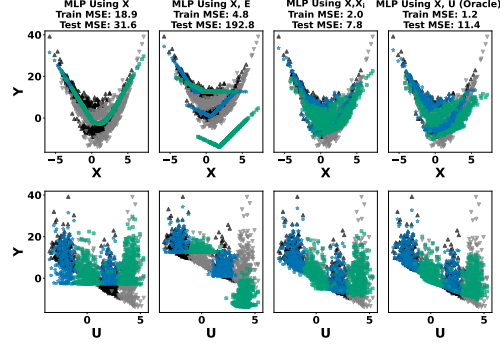


Figure D18: MLP performance in the low-overlap nonlinear setting, across conditioning sets: (i) X ; (ii) X and E (environment statistics); (iii) X and X_i ; (iv) X and U (oracle).

Structural Equations :

(Hidden Confounder) $U \sim \mathcal{N}(\mu_e, 0.5^2)$
 $\mu_e \in \{-1, 0\}$ for ID
 $\mu_e \in \{-0.5, 0.5\}$ for OOD
(Input) $X = 0.75U + \mathcal{N}(0, 1.5^2)$
(Outcome) $Y = X^2 - 3U + \mathcal{N}(0, 0.5^2)$
(Informative) $X_i = 0.6U + \mathcal{N}(0, 0.3^2)$

▲ Train data
▼ Test data
★ Predictions on train data
✱ Predictions on test data

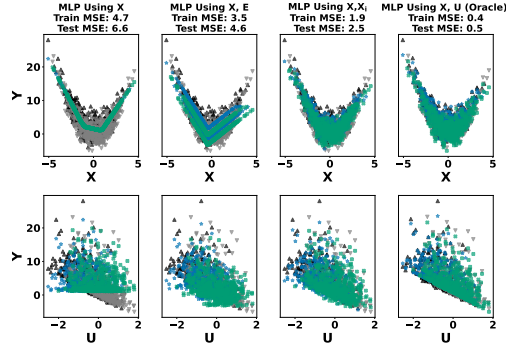


Figure D19: MLP performance in the high-overlap nonlinear setting, across conditioning sets: (i) X ; (ii) X and E (environment statistics); (iii) X and X_i ; (iv) X and U (oracle).

854 separate environments. Conditioning additionally on environment-specific summary statistics of the
855 observed covariates recovers part of the environment-specific structure, but the largest gains arise
856 only after appending informative covariates. For completeness we include the oracle estimator and
857 U - Y scatter plots that expose the learned input-output mappings. As expected, accuracy is higher
858 under high overlap; more notably, the estimators remain effective in the low-overlap regime—an
859 untested setting that violates the common-support assumption of prior work.

860 **Ratio of causal to anti-causal variables:** Recent benchmarks [24] report that baseline estimators
861 (e.g. XGB, MLP) match or exceed dedicated OOD generalization methods (e.g. GDRO, IRM, VREX)
862 on real-world tabular data, coinciding with the predominance of the structure $\mathbf{X} \rightarrow Y$. To determine
863 whether this baseline advantage is attributable to that structure, we sweep the causal-to-anti-causal

Structural Equations :

(Hidden Confounder) $U \sim \mathcal{N}(\mu_e, 0.5^2)$
 $\mu_e \in \{-2, 2\}$ for ID
 $\mu_e \in \{0, 4\}$ for OOD
(Input) $X = 0.75U + \mathcal{N}(0, 1.5^2)$
(Outcome) $Y = X^2 - 3U + \mathcal{N}(0, 0.5^2)$
(Informative) $X_i = 0.6U + \mathcal{N}(0, 0.3^2)$

▲ Train data
▼ Test data
★ Predictions on train data
✱ Predictions on test data

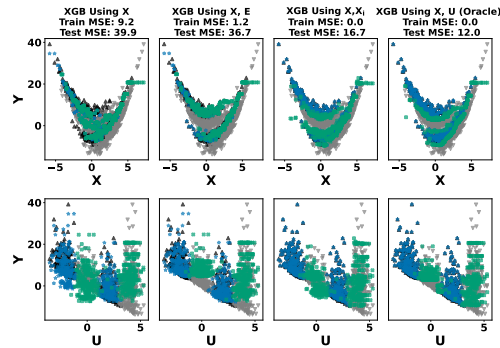


Figure D20: XGB performance in the low-overlap nonlinear setting, across conditioning sets: (i) X ; (ii) X and E (environment statistics); (iii) X and X_i ; (iv) X and U (oracle).

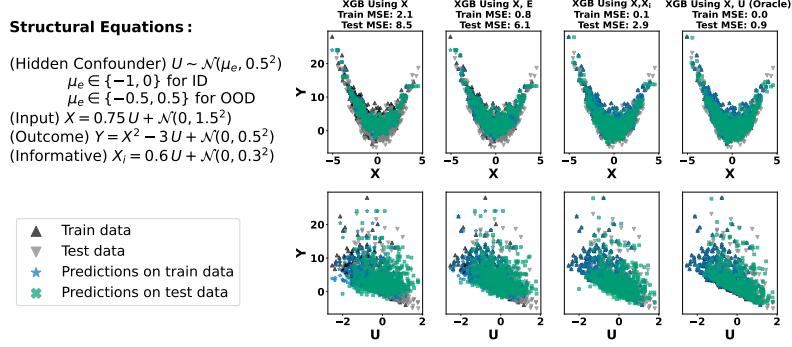


Figure D21: XGB performance in the high-overlap nonlinear setting, across conditioning sets: (i) X ; (ii) X and E (environment statistics); (iii) X and X_i ; (iv) X and U (oracle).

ratio $\rho = |\mathbf{X}_C|/|\mathbf{X}_A| \in \{0.0, \dots, 1.0\}$, fixing $|\mathbf{X}| = 50$ with $\mathbf{X} = \mathbf{X}_C \cup \mathbf{X}_A$ (so $\rho = 0.5$ gives $|\mathbf{X}_C| = |\mathbf{X}_A| = 25$). The baselines remain competitive even under the mixed structure $\mathbf{X}_C \rightarrow Y \rightarrow \mathbf{X}_A$ (Figure D22), although no single estimator dominates at every ρ (Figures D23, D24); the baselines attain the highest PMA-OD score [24] (OOD accuracy as a fraction of the per-setting maximum). The evaluated estimators are grouped as: (1) ERM-based (XGBoost [15], LightGBM [34], MLP, ResNet [27], SAINT [76], TabTransformer, NODE [61], FT-Transformer, ExpGrad [2]); (2) OOD generalization (IRM [6], IB-IRM, IB-ERM [30], CausIRL [17], DANN [3], MMD [43], DeepCORAL [81], V-REx [38], Domain Mixup [90, 91]); and (3) robust optimization (DRO [42], GroupDRO [69], Label DRO, Adversarial Label DRO [94]).

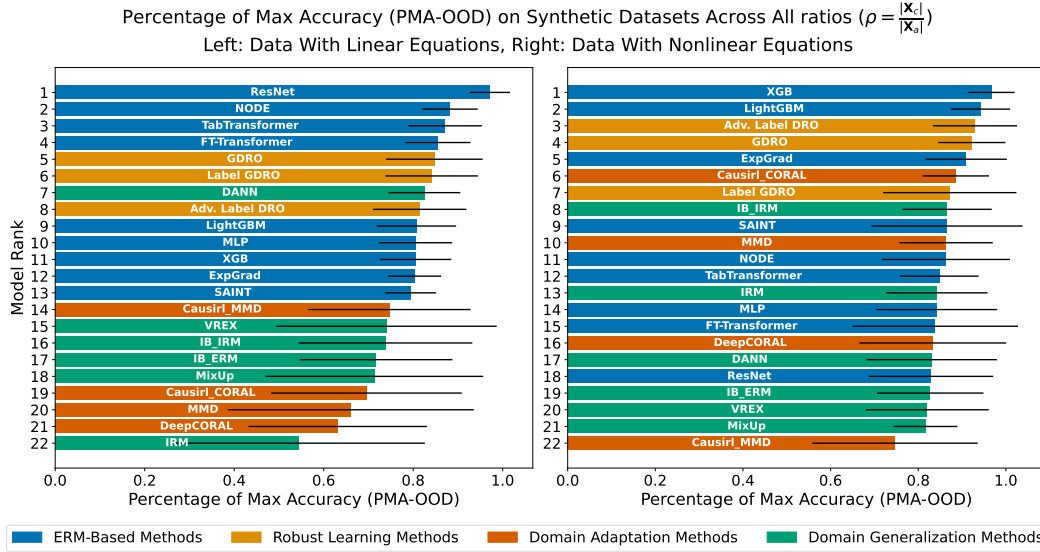


Figure D22: On the synthetic datasets, ERM-based estimators attain the highest percentage-of-maximum OOD (PMA-OD) test accuracy, reproducing the real-world ranking of Gardner et al. [24].

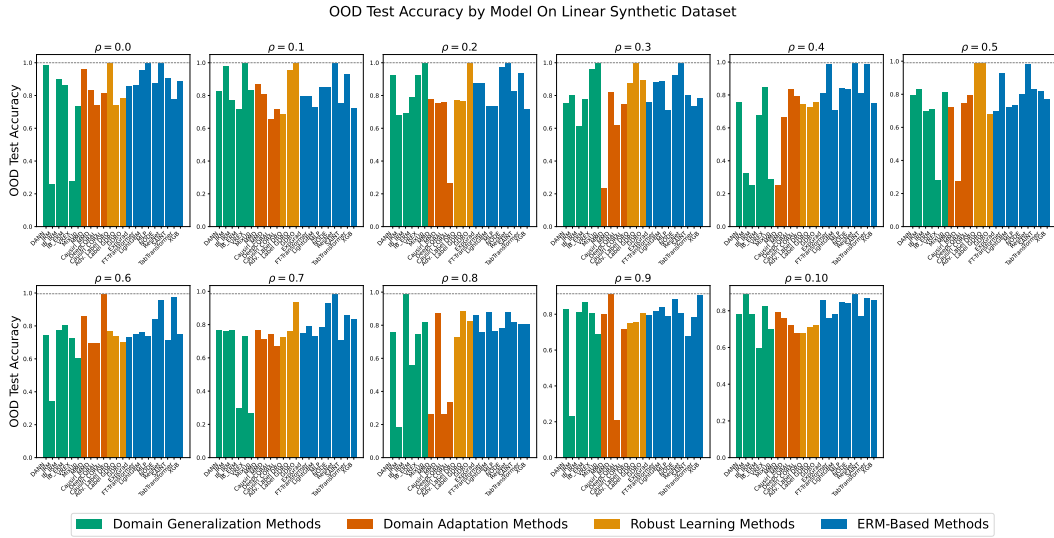


Figure D23: Per-estimator OOD test accuracy in the linear setting as a function of the causal-to-anti-causal ratio ρ .

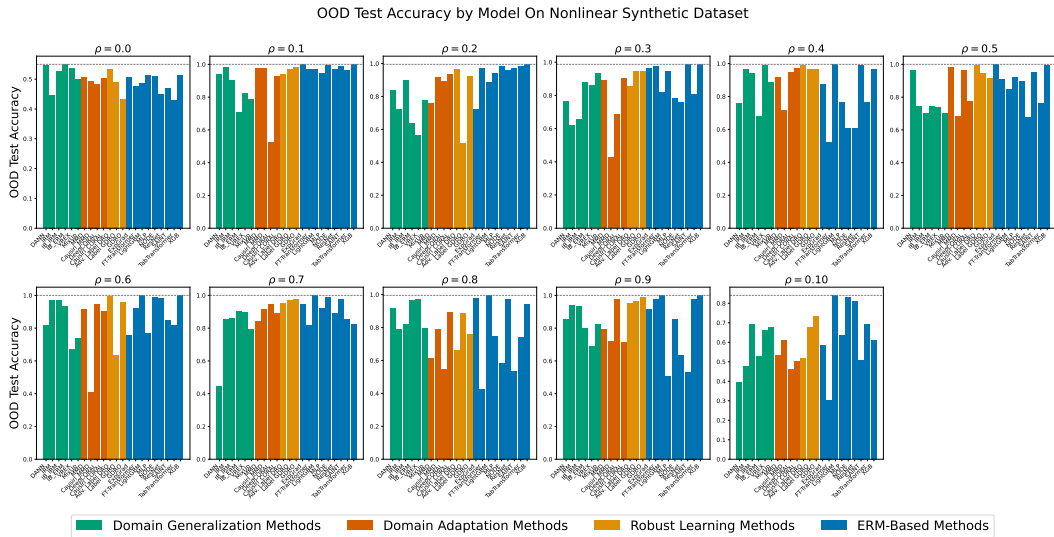


Figure D24: Per-estimator OOD test accuracy in the nonlinear setting as a function of the causal-to-anti-causal ratio ρ .

873 E Qualitative analysis of hidden confounding shift in real-world datasets

874 This section provides a qualitative assessment of hidden confounding shift in the real-world datasets.
875 Tables E3–E18 enumerate candidate hidden confounders acting on the observed covariates of several
876 benchmarks [57]; dataset descriptions are given in Gardner et al. [24], Nastl and Hardt [57]. The
877 candidates are elicited from GPT4o [1], and the resulting lists are indicative rather than exhaustive.
878 This evidence complements the quantitative diagnostics and supports the presence and cross-domain
879 impact of hidden confounding shift.

880 **Prompt:** The following prompt is issued to GPT4o to elicit candidate hidden confounders.

881 For a target variable, you will be given lists of causal, arguably
882 causal, anti-causal, and spurious covariates. Also, you will be provided
883 with environment variables such that different datasets are collected in
884 environments induced by these environment variables. Provide possible
885 hidden confounders that cause any of the target, causal, arguably causal,
886 spurious, and anti-causal covariates. That is, the hidden confounders
887 should influence the distribution of their children in different domains.
888 Ensure hidden confounders are not a part of the given variables, and your
889 explanation should include how hidden confounders influence their children
890 in different environments.

Table E3: **Dataset:** Food stamps. **Target:** Food stamp reciprocity in the past year for households with child. **Environments:** regions in the United States.

Hidden confounder	Affected variables	Reason for confounding
Local economic conditions	Food stamp reciprocity, Age, Sex, Race, Marital status	Poor economic conditions can increase food insecurity (raising food stamp usage), while also influencing population demographics due to migration, employment patterns, and household formation.
State or regional public assistance policies	Food stamp reciprocity, Marital status, Number of children, Household income	Generous or restrictive assistance policies directly affect food stamp eligibility and indirectly influence household structure and income distribution.
Cultural attitudes towards welfare and public assistance	Food stamp reciprocity, Marital status, Disability, Race, Ethnicity	Social stigma or support for welfare affects both the uptake of food stamps and social norms around marriage, disability reporting, and identity categories.
Regional health care conditions	Food stamp reciprocity, Disability, Cognitive difficulty, Hearing difficulty, Vision difficulty	Regions with limited healthcare access may show higher disability prevalence and greater reliance on food stamps due to increased financial and care burdens.

Table E4: **Dataset:** Income. **Target:** Total person’s income $\geq 56K$ for employed adults. **Environments:** regions in the United States.

Hidden confounder	Affected variables	Reason for confounding
Local economic conditions	Income, Occupation, Educational attainment, Marital status	Stronger local economies offer better-paying jobs and education access, increasing income; they also influence occupational choices and household formation patterns.
State or regional tax and labor policies	Income, Class of worker, Educational attainment, Marital status	Tax incentives and labor protections affect wages and employment types, while also shaping decisions about education and family due to financial security.
Cost of living and regional affordability	Income, Occupation, Marital status, Educational attainment	Higher living costs necessitate higher incomes and may drive occupational or educational shifts; they also influence marriage or cohabitation decisions.
Cultural norms and regional economic history	Income, Educational attainment, Occupation	Cultural values and historical industry presence shape education levels and job opportunities, which in turn affect income distributions.

Table E5: **Dataset:** Public coverage. **Target:** Coverage of non-Medicare eligible low-income individuals. **Environments:** Disability statuses.

Hidden confounder	Affected variables	Reason for confounding
State or regional healthcare policies	Public health coverage, Marital status, Employment status, Income	Differences in Medicaid expansion and public insurance eligibility influence health coverage; these policies also affect economic stability, employment, and family dynamics.
Economic conditions and poverty levels	Public health coverage, Employment status, Income, Marital status	Poorer regions tend to have lower employment rates and incomes, which increase reliance on public health coverage and influence marriage and household composition.
Cultural attitudes towards disability and healthcare	Public health coverage, Disability status, Cognitive difficulty, Hearing difficulty, Vision difficulty	Stigma or support for disability and public care varies culturally, affecting both the reporting of disabilities and the likelihood of seeking or receiving public coverage.
Urban vs. rural divide	Public health coverage, Educational attainment, Marital status, Occupation	Urban areas typically offer better healthcare access, education, and jobs, all of which affect coverage likelihood and socioeconomic indicators.

Table E6: **Dataset:** Unemployment. **Target:** Classify whether a person is unemployed. **Environments:** Educational attainments.

Hidden confounder	Affected variables	Reason for confounding
Local economic conditions	Employment status, Occupation, Marital status, Mobility status	Regional economic strength affects job availability and unemployment rates; it also shapes occupation types, migration decisions, and household stability.
State or regional labor market policies	Employment status, Occupation, Educational attainment, Marital status	Labor regulations and unemployment benefits vary by region, affecting hiring practices, education incentives, and family structures.
Cultural and social norms around employment	Employment status, Disability status, Marital status	Cultural attitudes toward work and dependency influence unemployment reporting and societal roles around disability and family responsibilities.
Health and disability status	Employment status, Disability status, Cognitive difficulty, Hearing difficulty, Vision difficulty	Poor health or disabilities reduce the ability to work, directly increasing unemployment and shaping associated health-related variables.

Table E7: **Dataset:** Voting. **Target:** Classify whether a person voted in the U.S. presidential election. **Environments:** United States census regions.

Hidden confounder	Affected variables	Reason for confounding
Political, family, and peer influences	Voted in national election, Party identification, Political participation	Social networks shape political ideology and engagement, influencing both voting likelihood and party alignment.
Media consumption habits	Voted in national election, Party identification, Political knowledge, Voting behavior	Media exposure affects awareness of political issues and biases, influencing party affiliation, political knowledge, and voting participation.
Social capital	Voted in national election, Political participation, Interest in elections	Strong community ties and civic networks increase political interest and participation, leading to higher voter turnout.
Civic education and political engagement programs	Voted in national election, Interest in elections, Political knowledge	Educational programs raise political awareness and civic responsibility, influencing both knowledge levels and voting decisions.
Historical and cultural context	Voted in national election, Party identification, Interest in elections	Historical events and regional political culture affect interest in elections and party alignment, which in turn influence voting behavior.

Table E8: **Dataset:** Hypertension. **Target:** Whether a person has hypertension. **Environments:** Body Mass Index (BMI) values.

Hidden confounder	Affected variables	Reason for confounding
Socioeconomic status	Income, Employment status, Smoking habits, Alcohol consumption, Physical activity, Healthcare access, Medical cost, High blood pressure diagnosis	Lower socioeconomic status reduces access to healthcare and healthy lifestyle options, leading to poor diet, limited activity, and delayed diagnosis, which jointly influence both BMI and hypertension risk.
Access to healthcare services	Healthcare access, Medical costs, Smoking habits, Alcohol consumption, Physical activity, High blood pressure diagnosis	Limited healthcare access results in underdiagnosis and unmanaged hypertension, while also affecting lifestyle choices that vary with BMI, confounding the relationship between BMI and hypertension.
Genetic predisposition	Age group, Race, Sex, Smoking habits, Diabetes, High blood pressure diagnosis	Genetic risk factors for hypertension may co-vary with demographic attributes and influence both hypertension prevalence and BMI distribution across subpopulations.
Psychosocial stress	Smoking habits, Alcohol consumption, Physical activity, High blood pressure diagnosis, Diabetes, Age group	Chronic stress alters behavior (e.g., smoking, inactivity) and physiological responses, contributing to both increased BMI and elevated blood pressure, thus confounding the BMI–hypertension link.
Environmental factors	Physical activity, Diet, Smoking habits, High blood pressure diagnosis, Diabetes, BMI category	Living environments affect access to recreational spaces, food quality, and pollution exposure, influencing both BMI and hypertension risks. These vary across BMI categories, creating confounding.
Dietary habits beyond fruits and vegetables	Alcohol consumption, Smoking habits, Physical activity, High blood pressure diagnosis, BMI category	High-sodium or processed-food diets raise both BMI and hypertension risk. Variation in such unmeasured dietary habits across BMI categories creates spurious associations with hypertension.

Table E9: **Dataset:** College Scorecard. **Target:** Predict completion rate for first-time, full-time students at four-year institutions. **Environments:** Based on Carnegie Classifications.

Hidden confounder	Affected variables	Reason for confounding
Institutional funding and re-sources	Accreditor, Control of institution, Highest degree awarded, In-state tuition, Out-of-state tuition, Cost of attendance, SAT scores	Wealthier institutions can offer better academic support, facilities, and programs, leading to higher completion rates and more selective admission profiles. This varies across Carnegie classifications.
Regional socio-economic factors	Region, Poverty rate, Unemployment rate, SAT scores	Economic conditions across regions affect affordability, student preparedness, and institutional support levels, all influencing both enrollment outcomes and graduation likelihood.
Demographic factors	HBCU flag, Federal loan recipient rate, Pell grant recipient rate, ACT scores, Undergraduate enrollment	Student demographics shape financial aid needs, academic preparation, and graduation rates. Their effect differs by institution type and selectivity under Carnegie categories.
Community support and engagement	Distance-education flag, Federal loan recipient rate, Pell grant recipient rate, SAT scores, Undergraduate enrollment	Supportive institutional communities improve retention and completion. Variation in engagement across institution types and student aid profiles induces confounding.
Admission selectivity	Admission rate, SAT (reading/math) midpoints, ACT midpoint, Undergraduate enrollment	Selective admissions correlate with better-prepared students and higher completion rates, and vary with institutional prestige and classification.
State and local policies	Region, Poverty rate, Unemployment rate, Cost of attendance	Differences in education funding and public policy affect cost structures and completion outcomes, interacting with institutional classification and regional demographics.

Table E10: **Dataset:** ASSISTments. **Target:** Predict whether a student solves a problem correctly on the first attempt in an online learning tool. **Environments:** Different schools.

Hidden confounder	Affected variables	Reason for confounding
Institutional teaching quality	Hint count, Attempt count, Skill ID, Problem type, Tutor mode, Position, Type, First action, Milliseconds to first response, Overlap time, Average confidence	Variation in instructional quality and pedagogy across schools affects how effectively students engage with content, leading to differences in problem-solving strategies, response behavior, and emotional states.
Student motivation	Hint count, Attempt count, Skill ID, Problem type, First action, Milliseconds to first response, Overlap time, Average confidence	Differences in intrinsic motivation across schools influence students' willingness to persevere, seek help, or give up quickly, affecting interaction and performance.
Classroom environment	Hint count, Attempt count, Tutor mode, Position, Type, First action, Average confidence	Peer dynamics, classroom culture, and noise levels vary by school and affect how confidently and independently students solve problems.
School technology infrastructure	Hint count, Attempt count, Tutor mode, Position, Type, First action, Milliseconds to first response, Overlap time, Average confidence	Access to reliable devices and fast internet differs by school, influencing response time, tool usage, and student experience.
Teacher-student interaction	Hint count, Attempt count, Tutor mode, Position, Type, First action, Milliseconds to first response, Overlap time, Average confidence	The level of teacher guidance and feedback shapes how much support students require during problem-solving, affecting engagement and confidence.
Previous academic performance	Hint count, Attempt count, Skill ID, Problem type, First action, Average confidence	Students' prior achievement affects how easily they solve problems, their need for assistance, and their confidence, all of which vary across schools.

Table E11: **Dataset:** ICU. **Target:** Predict whether the patient will stay in the ICU for longer than 3 days. **Environments:** Insurance types.

Hidden confounder	Affected variables	Reason for confounding
Socioeconomic status (SES)	Age, Gender, Ethnicity, Height, Weight, Bicarbonate, CO ₂ , pCO ₂ , pO ₂ , Lactate, Sodium, Hemoglobin, Oxygen saturation, Respiratory rate, etc.	SES influences access to healthcare, preventive services, and overall health status. Differences in SES across insurance types lead to variability in pre-ICU health, physiological indicators, and ICU stay duration.
Hospital resources and care quality	Age, Gender, Ethnicity, Height, Weight, Bicarbonate, CO ₂ , pCO ₂ , pO ₂ , Lactate, Sodium, Hemoglobin, Oxygen saturation, Respiratory rate, Heart rate, etc.	Hospital infrastructure and care standards affect monitoring, intervention speed, and clinical decisions. These factors vary by insurance coverage and influence ICU outcomes and vitals.
Comorbidities	Age, Gender, Ethnicity, Height, Weight, Bicarbonate, CO ₂ , pCO ₂ , pO ₂ , Lactate, Sodium, Hemoglobin, Oxygen saturation, Respiratory rate, Heart rate, etc.	Presence of chronic conditions (e.g., diabetes, cardiovascular disease) affects both the need for prolonged ICU care and physiological measurements. The distribution of comorbidities differs across insurance types.
Insurance-related treatment variability	Age, Gender, Ethnicity, Height, Weight, Bicarbonate, CO ₂ , pCO ₂ , pO ₂ , Lactate, Sodium, Hemoglobin, Oxygen saturation, Respiratory rate, etc.	Differences in treatment timing, intensity, and access to specialists based on insurance policies affect ICU stay duration and clinical metrics.
Genetic factors	Age, Gender, Ethnicity, Height, Weight, Bicarbonate, CO ₂ , pCO ₂ , pO ₂ , Lactate, Sodium, Hemoglobin, etc.	Inherited traits influence predisposition to organ failure, metabolic responses, and recovery trajectories. These effects are partially mediated by ethnicity and age distributions, which vary across insurance groups.

Table E12: **Dataset:** Hospital mortality. **Target:** Classify whether an ICU patient expires in the hospital during their current visit. **Environments:** Insurance types.

Hidden confounder	Affected variables	Reason for confounding
Socioeconomic status (SES)	Age, Gender, Ethnicity, Height, Weight, Bicarbonate, Lactate, Sodium, Hemoglobin, Oxygen saturation, Respiratory rate, Systolic blood pressure, White blood cell count, etc.	SES shapes access to timely and high-quality care, preventive services, and general health status. Patients with higher SES often have better insurance and outcomes, leading to confounding with mortality risk.
Comorbidities	Age, Gender, Ethnicity, Height, Weight, Bicarbonate, CO ₂ , pCO ₂ , pO ₂ , Lactate, Sodium, Hemoglobin, Oxygen saturation, Respiratory rate, Heart rate, Systolic blood pressure, etc.	Pre-existing conditions such as diabetes or heart disease increase mortality risk and influence physiological features. Their prevalence differs by insurance type, creating confounding.
Hospital resources and care quality	Age, Gender, Ethnicity, Height, Weight, Bicarbonate, CO ₂ , pCO ₂ , pO ₂ , Lactate, Sodium, Hemoglobin, Oxygen saturation, Respiratory rate, Heart rate, Systolic blood pressure, White blood cell count, etc.	Access to advanced treatments, trained staff, and timely interventions influences survival rates. These factors correlate with insurance coverage, confounding mortality outcomes.
Genetic factors	Age, Gender, Ethnicity, Height, Weight, Bicarbonate, CO ₂ , pCO ₂ , pO ₂ , Lactate, Sodium, Hemoglobin, Oxygen saturation, Respiratory rate, Heart rate, etc.	Genetic predispositions affect disease susceptibility and treatment responses. Variations in genetic risk factors may correlate with demographic traits across insurance types.
Lifestyle and behavioral factors	Age, Gender, Ethnicity, Height, Weight, Bicarbonate, CO ₂ , pCO ₂ , pO ₂ , Lactate, Sodium, Hemoglobin, Oxygen saturation, Respiratory rate, Heart rate, Temperature, Systolic blood pressure, White blood cell count, etc.	Behaviors such as smoking, diet, and physical activity affect long-term health and mortality risk. These behaviors vary systematically with SES and insurance coverage, influencing both target and physiological features.

Table E13: **Dataset:** Childhood lead. **Target:** Predict blood lead levels above CDC blood level reference value. **Environments:** Poverty-income ratios.

Hidden confounder	Affected variables	Reason for confounding
Environmental Exposure	Blood lead levels, Country of birth, Age, Race and Hispanic origin	Environmental exposure to lead influences blood lead levels, and this varies significantly across socio-economic groups. People in lower PIR groups are more likely to live in areas with higher lead contamination, which contributes to higher blood lead levels. Furthermore, environmental factors may affect the demographic distribution (e.g., country of birth, race).
Access to Healthcare	Blood lead levels, Age, Gender, Race and Hispanic origin, Marital status, Education	Limited access to healthcare, especially in lower PIR groups, means fewer opportunities for detection and treatment of lead poisoning. This results in higher blood lead levels, with disparities also influencing demographic variables like age, gender, and education. Additionally, healthcare access varies by insurance and socio-economic status, further confounding the relationships.
Diet and Nutrition	Blood lead levels, Age, Gender, Race and Hispanic origin	Dietary factors, such as poor nutrition in lower PIR groups, can exacerbate lead absorption. Malnutrition increases the body's susceptibility to lead poisoning, raising blood lead levels. In contrast, higher PIR groups may have better access to nutritious foods, lowering lead absorption, thus creating a confounding effect in how socio-economic status and race influence lead toxicity.
Housing Conditions	Blood lead levels, Country of birth, Race and Hispanic origin, Marital status	Older housing conditions, which are more prevalent in lower PIR groups, contribute significantly to elevated lead exposure (e.g., lead paint, poor plumbing). These living conditions directly influence blood lead levels and can also correlate with demographic factors like country of birth, race, and marital status. This introduces confounding, as socio-economic status impacts both exposure and the demographics of affected individuals.
Occupation	Blood lead levels, Age, Race and Hispanic origin, Marital status, Education	Certain occupations, which are more common among lower PIR groups, involve higher lead exposure (e.g., construction, manufacturing). Occupational lead exposure directly impacts blood lead levels and is often correlated with education, marital status, and socio-economic status. The varying prevalence of lead exposure by occupation introduces confounding, especially across different PIR groups.

Table E14: **Dataset:** Diabetes. **Target:** Predict diabetes. **Environments:** Preferred race categories.

Hidden confounder	Affected variables	Reason for confounding
Genetic predisposition	Diabetes, BMI, High blood pressure, High blood cholesterol	Genetic factors and family history contribute to both the onset of diabetes and comorbid conditions like obesity, hypertension, and high cholesterol. These genetic predispositions can make individuals more susceptible to diabetes, leading to confounding as they correlate with other health indicators.
Access to healthcare	Diabetes, Physical health, BMI, Healthcare coverage, Health checkups	Limited or unequal access to healthcare, especially in marginalized racial groups, leads to disparities in diabetes diagnosis, management, and comorbidity treatment. It also influences the frequency of health checkups and access to medications, which can confound the relationship between diabetes status and other health metrics.
Dietary habits and food availability	Diabetes, BMI, Physical health, Alcohol consumption, Fruit and vegetable intake	Dietary habits, often shaped by socioeconomic status and local food environments, influence weight, health behaviors (such as alcohol consumption), and diabetes risk. People in lower socioeconomic strata may have limited access to healthy food options, leading to higher BMI and increased diabetes risk, creating confounding effects on health outcomes.
Psychosocial stress and mental health factors	Diabetes, Mental health, Physical health, BMI, Physical activity, Doctor visits	Chronic stress and mental health issues, often higher in marginalized groups, contribute to diabetes development and complicate its management. These factors also affect physical health (e.g., weight gain due to stress) and health-seeking behaviors (e.g., fewer doctor visits), leading to confounding by influencing both diabetes risk and its associated variables.
Socioeconomic status beyond income	Diabetes, Income, Physical health, Healthcare coverage, Education level	Socioeconomic factors, such as occupation, education, and neighborhood wealth, influence access to healthcare, nutrition, and overall health behaviors. These factors can confound the relationships between diabetes and other socio-economic variables like income and education, as they shape opportunities for prevention and treatment.

Table E15: **Dataset:** Sepsis. **Target:** Predict, from a set of fine-grained ICU data, whether a patient will experience sepsis onset within the next 6 hours. **Environments:** Lengths of ICU stay.

Hidden confounder	Affected variables	Reason for confounding
Infection prevalence in ICU	SepsisLabel, Temperature (Temp), Leukocyte count (WBC), Heart rate (HR), Blood urea nitrogen (BUN)	Higher infection rates in certain ICU units can lead to a higher probability of sepsis onset (SepsisLabel). These infection rates influence biomarkers such as WBC, HR, and BUN, creating confounding because the unit's infection environment affects both the likelihood of sepsis and the observed clinical measures.
Quality of ICU Care	SepsisLabel, Fibrinogen concentration (Fibrinogen), Leukocyte count (WBC), Platelet count (Platelets)	Higher-quality care in certain ICUs may lead to earlier identification and treatment of sepsis, resulting in more accurate SepsisLabel predictions. Additionally, better care could affect biomarkers like fibrinogen, WBC, and platelets, which are critical in sepsis detection and progression, thereby confounding the relationships between these variables and the outcome.
Severity of underlying conditions	SepsisLabel, Blood urea nitrogen (BUN), Creatinine, Lactate, Calcium	Patients with severe chronic conditions (e.g., kidney disease, cardiovascular issues) are at a higher risk of sepsis and may show abnormal levels in biomarkers like BUN, creatinine, lactate, and calcium. These underlying conditions contribute to the SepsisLabel outcome and confound the relationship between the biomarkers and the likelihood of sepsis, varying across ICU units depending on patient population.
Patient's socio-economic status	Age (Age), Gender (Gender), Leukocyte count (WBC), Fibrinogen concentration (Fibrinogen), Platelet count (Platelets)	Socio-economic factors, such as access to healthcare, can influence both the likelihood of sepsis and the observed clinical biomarkers. For example, patients from lower socio-economic backgrounds may have delayed hospitalizations or inadequate care, which affects both SepsisLabel and the progression of sepsis as indicated by WBC, fibrinogen, and platelet levels.
Hospital-specific protocols and treatment guidelines	SepsisLabel, Lactate, Glucose, Creatinine	Differences in hospital protocols for sepsis treatment, such as timing of interventions and choice of sepsis bundles, can affect both the SepsisLabel and biomarkers like lactate, glucose, and creatinine. These protocols lead to variability in how sepsis is diagnosed and treated across different hospitals and ICU units, confounding the relationship between biomarkers and sepsis outcomes.

Table E16: **Dataset:** Hospital readmission. **Target:** Predict whether a diabetic patient is readmitted to the hospital within 30 days of their initial release. **Environments:** Admission sources.

Hidden confounder	Affected variables	Reason for confounding
Socio-economic status (SES)	Race, Gender, Age, Payer code, Medical specialty, Number of outpatient visits, Number of emergency visits, Number of inpatient visits, Diabetes medication prescribed	SES influences access to healthcare, patient demographics, and chronic disease rates. It can also determine the type of care received based on admission source (e.g., emergency department vs. outpatient settings). Differences in healthcare access, such as availability of medications, may impact readmission rates and associated variables like outpatient visits and prescribed medication.
Severity of illness	Primary diagnosis, Secondary diagnosis, Number of diagnoses, Discharge type, Medication changes (e.g., Insulin, Glipizide)	More severe illness increases the likelihood of readmission and influences the complexity of diagnoses and the treatments administered. The severity of illness may differ based on the admission source (e.g., emergency versus outpatient), impacting the number and type of diagnoses and treatments prescribed at discharge, affecting readmission likelihood.
Access to healthcare resources	Time in hospital, Discharge disposition, Number of procedures, Number of medications, Number of lab tests	Access to healthcare resources (e.g., time in hospital, availability of procedures and medications) influences treatment decisions and outcomes. Different admission sources may have varying levels of available resources, leading to different lengths of stay, the types of procedures performed, and overall treatment quality, which can affect the likelihood of readmission.
Patient's adherence to medication	Change in medications, Diabetes medication prescribed, Number of outpatient visits, Number of emergency visits	Adherence to prescribed medications is often influenced by socio-economic status, which can vary depending on admission source. Non-adherence may lead to medication changes, affecting diabetes control and subsequent readmission risk. The number of outpatient and emergency visits can also reflect how well a patient manages their diabetes and the likelihood of complications.
Hospital-specific protocols	Time in hospital, Number of diagnoses, Discharge disposition, Readmitted	Different hospitals and healthcare systems implement various protocols for discharge planning and readmission prevention, which can affect readmission rates. These protocols may vary by admission source, where patients admitted via the emergency department may receive different follow-up instructions and care than those admitted through other channels.

Table E17: **Dataset:** MEPS. **Target:** Measure of health care utilization. **Environments:** Insurance types.

Hidden confounder	Affected variables	Reason for confounding
Socioeconomic status (SES)	Years of education, Employment status, Hourly wage, Paid sick leave, Paid leave to visit doctor, Family size, Insurance coverage, Healthcare utilization	SES can influence access to healthcare services, insurance coverage, and the ability to use medical services. Insurance types often correlate with SES levels, where individuals with lower SES may be more likely to be insured by government programs (e.g., Medicaid), which in turn impacts healthcare utilization patterns across different SES groups.
Healthcare access	Paid sick leave, Insurance coverage, Employer offers health insurance, Healthcare utilization	Limited access to healthcare, such as lack of insurance or paid sick leave, directly impacts healthcare utilization. The type of insurance a person holds is often tied to access to various medical services. The level of coverage and accessibility differs across insurance types, influencing healthcare behaviors such as whether a patient can afford and utilize healthcare services.
Health behaviors	Perceived health status, Asthma medications, Limitations in physical functioning, Healthcare utilization	Lifestyle factors like smoking, alcohol consumption, and physical activity can directly affect health status and healthcare needs. Health behaviors differ across groups with different insurance types, and these behaviors contribute to healthcare utilization. Insurance coverage can also be influenced by perceived health status, which varies across insured groups, affecting utilization of medical services.
Chronic health conditions	Asthma medications, Perceived health status, Limitations in physical functioning, Healthcare utilization	Chronic conditions like asthma or diabetes increase the need for healthcare services, leading to higher healthcare utilization. People with chronic conditions are more likely to be covered by Medicare or Medicaid, which influences the type of insurance they hold and impacts their healthcare utilization and associated features (e.g., medications and physical limitations).
Regional healthcare infrastructure	Region, Family size, Healthcare utilization, Paid sick leave, Paid leave to visit doctor, Employment status	The quality and availability of healthcare infrastructure vary across regions, which can impact healthcare utilization. In underserved regions, people may be more reliant on public insurance options like Medicaid, affecting healthcare access and behaviors. The regional differences in healthcare systems can lead to disparities in access to medical services, insurance coverage, and utilization of healthcare.

Table E18: **Dataset:** Poverty. **Target:** Predict household income-to-poverty ratio. **Environments:** Citizenship statuses.

Hidden confounder	Affected variables	Reason for confounding
Social capital	Income, Unemployment compensation, Disability benefits, Family size, Housing conditions, Household income-to-poverty ratio	Stronger social support networks can improve access to income, benefits, and resources like unemployment or disability compensation. Social capital can vary based on citizenship status, which in turn influences income, housing conditions, and eligibility for benefits, confounding the relationship between household income and poverty ratios.
Workplace discrimination or bias	Household income, Worker's compensation, Unemployment compensation, Pension, Family income, Disability benefits	Discrimination in the workplace can limit opportunities for higher wages, pensions, or disability benefits, especially for marginalized groups (e.g., racial minorities, non-citizens). This bias varies by citizenship status and leads to biased associations between income, benefits, and poverty levels.
Healthcare access	Household income, Health insurance premiums, Medical expenses, Disability benefits, Family size, Health conditions, Medicaid/Medicare assistance	Limited access to healthcare, particularly for non-citizens, can lead to higher medical expenses and greater reliance on public health assistance. Citizenship status directly impacts eligibility for public healthcare programs, confounding relationships between medical expenses and income, and affecting poverty ratios.
Access to education and skills development	Educational attainment, Household income, Employment status, Unemployment compensation, Income from assistance, Savings	Disparities in educational opportunities, often linked to citizenship status, influence income potential and eligibility for government assistance. These disparities, in turn, affect the household income-to-poverty ratio and employment outcomes, creating confounding relationships between education and income.
Cultural factors	Family size, Household income, Savings, Disability benefits, Healthcare utilization, Income assistance, Living arrangements	Cultural factors influence financial management, family support, and the use of social programs. These factors can vary across citizenship statuses, leading to differences in how household income and assistance are distributed, and thus confounding the relationship between income, benefits, and financial behavior.
Housing market and rent conditions	Housing ownership, Housing costs, Family size, Household income, Rent payments, Social security benefits, Medical aid	Housing market conditions, especially rent disparities, can create financial strain and impact the household income-to-poverty ratio. Citizenship status often influences eligibility for housing assistance and rent subsidies, which confounds the relationship between housing costs and income, particularly in areas with significant immigrant populations.